Bloch

Malcolm

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Contents

Chapter 1

Construction of the Real Numbers

Definition 1.1. *Let S be a set. A* ***binary operation*** *on S is a function $S \times S \to S$. A* ***unary operation*** *on S is a function $S \to S$.*

Let S be a set and let $* : S \times S \to S$ be a binary operation. If $x, y \in S$, the correct way to write the result of the operation to the pair $\langle x, y\rangle$ would be $*(\langle x, y\rangle)$. However since this is a bit cumbersome one may write $x * y$ instead. Similarly if $\neg : S \to S$ is a unary operation and $x \in S$ one may choose to write $\neg x$ over $\neg(x)$.

1.1 Axioms for the Natural numbers

1.1.1 Peano Postulates

Axiom: Peano Postulates
There exists a set $\mathbb{N}$ with an element $1 \in \mathbb{N}$ and a function $s : \mathbb{N} \to \mathbb{N}$ that satisfy the following three properties.

- There is no $n \in \mathbb{N}$ such that $s(n) = 1$.
- The function s is injective.
- Let $G \subseteq \mathbb{N}$ be a set. Suppose that $1 \in G$, and that if $g \in G$ then $s(g) \in G$. Then $G = \mathbb{N}$.

It will be shown that the set $\mathbb{N}$ is unique. We can therefore give it a name.

Definition 1.2. *The set of* ***natural numbers*** *denoted $\mathbb{N}$, is the set the existence of which is given in the Peano Postulates.*

1.1.2 Consequences of Peano Postulates

Lemma 1.3. *Let $a \in \mathbb{N}$. Suppose that $a \neq 1$. Then there is a unique $b \in \mathbb{N}$ such that $a = s(b)$. (where existence of s and $\mathbb{N}$ are outlined by the peano axioms)*

Proof. First we prove existence. Let

$$G = \{1\} \cup \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$$

(which exists by the subset, pairing, and union axioms) We will prove that $G = \mathbb{N}$. To do this we need to show $1 \in G$, $G \subseteq \mathbb{N}$ and $\forall g \in G(s(g) \in G)$. The first two are straightforward. For the third suppose $g \in G$; we have two cases. First suppose $g \in \{1\}$; then $g = 1 \in \mathbb{N}$. Since $s : \mathbb{N} \to \mathbb{N}$, we then have $s(1) \in \mathbb{N}$. So by definition $s(1) \in \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$ and $s(g) = s(1) \in G$. Now suppose $g \in \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$; then by definition $g \in \mathbb{N}$ and $s(g) \in \mathbb{N}$ exists. By definition $s(g) \in \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$ so $s(g) \in G$. By the third peano postulate we then can assert that $G = \mathbb{N}$. So $a \in \mathbb{N} = G$. Since $a \neq 1$ we must then have $a \in \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$ and by definition, $\exists b \in \mathbb{N}(s(b) = a)$.

Now for uniqueness. Let m and n be arbitrary elements of $\mathbb{N}$ such that $a = s(m)$ and $a = s(n)$. Then $\langle m, a\rangle \in s$ and $\langle n, a\rangle \in s$. Since s is injective we must have $n = m$. □

Theorem 1.4. *Let H be a set, Let $e \in H$ and let $k : H \to H$ be a function. Then there is a unique function $f : \mathbb{N} \to H$ such that $f(1) = e$ and $f \circ s = k \circ f$.*

Proof. Letting H, e, and k be arbitrary such that $e \in H$ and $k : H \to H$. We will show existence and uniqueness separately. We start with uniqueness.

Let g and h be arbitrary such that $g : \mathbb{N} \to H$, $g(1) = e$, and $g \circ s = k \circ g$ and $h : \mathbb{N} \to H$, $h(1) = e$, and $h \circ s = k \circ h$. We will first define V as the unique set (guaranteed by the subset and extensionality axioms) such that

$$\forall a(a \in V \leftrightarrow a \in \mathbb{N} \wedge g(a) = h(a))$$

and show that $V = \mathbb{N}$. By peano's postulates, we need to show $V \subseteq \mathbb{N}$, $1 \in V$, and $\forall n \in V(s(n) \in V)$. The first is straightforward. For the second requirement, we know by peano's postulates that $1 \in \mathbb{N}$, so by our assumptions $g(1)$ and $h(1)$ exist and $g(1) = e = h(1)$. So $1 \in V$.

For the third requirement, let n be arbitrary and suppose that $n \in V$. Then by definition $n \in \mathbb{N}$ and $g(n) = h(n)$. Since $n \in \mathbb{N} = \mathrm{dom}(h \circ s)$, we have, by our assumptions,

$$h(s(n)) = h \circ s(n) = k \circ h(n) = k(h(n)) = k(g(n)) = k \circ g(n) = g \circ s(n) = g(s(n))$$

we know by definition $\mathrm{ran}(s) \subseteq \mathbb{N}$, so $s(n) \in \mathbb{N}$ and again by definition $s(n) \in V$. We can conclude that $V = \mathbb{N}$.

Suppose $\langle x, y\rangle \in g$, then $x \in \mathrm{dom} g = \mathbb{N} = V$. So by definition $y = g(x) = h(x)$, and $\langle x, y\rangle \in h$. Suppose $\langle x, y\rangle \in h$, then $x \in \mathrm{dom} h = \mathbb{N} = V$. So by definition $y = h(x) = g(x)$, and $\langle x, y\rangle \in g$.
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Now for existence. We need to come up with a function from $\mathbb{N}$ into H such that $f(1) = e$ and $f \circ s = k \circ f$. We first define C, where

$$\forall X(X \in C \leftrightarrow X \subseteq \mathbb{N} \times H \wedge \langle 1, e\rangle \in X \wedge \forall n \forall y(\langle n, y\rangle \in X \rightarrow \langle s(n), k(y)\rangle \in X))$$

(guaranteed by the subset and extensionality axioms.) We propose that $f = \bigcap C$ is the desired function. To ensure its existence, we need to show that C is nonempty. We will show that $\mathbb{N} \times H \in C$. For the first requirement, clearly $\mathbb{N} \times H \subseteq \mathbb{N} \times H$. Next, since $1 \in \mathbb{N}$ and $e \in H$ then $\langle 1, e\rangle \in \mathbb{N} \times H$. Finally, let n and y be arbitrary such that $\langle n, y\rangle \in \mathbb{N} \times H$. Since $s : \mathbb{N} \rightarrow \mathbb{N}$ and $k : H \rightarrow H$, we must then have $s(n) \in \mathbb{N}$ and $k(y) \in H$. So $\langle s(n), k(y)\rangle \in \mathbb{N} \times H$. $f = \bigcap C$ is therefore a valid candidate.

Before we prove the functionality of f, we first state a few facts about f. First we note that $f \in C$. Let arbitrary $x \in \bigcap C$. Since C is nonempty there exists $X \in C$, and by definition $x \in X \subseteq \mathbb{N} \times H$. So $\bigcap C \subseteq \mathbb{N} \times H$. Next let arbitrary $X \in C$. By definition $\langle 1, e\rangle \in X$, so $\langle 1, e\rangle \in \bigcap C$. Finally let n and y be arbitrary and suppose $\langle n, y\rangle \in \bigcap C$. We need to show that $\langle s(n), k(y)\rangle \in \bigcap C$. Let arbitrary $X \in C$. We have $\langle n, y\rangle \in X$. Since $X \in C$, by definition we must then have $\langle s(n), k(y)\rangle \in X$. We are done.

Next we note that $\forall X \in C(f \subseteq X)$. This is straightforward to prove. Finally we show

$$\begin{aligned}\forall n \forall y([\langle n, y\rangle \in f \wedge \langle n, y\rangle \neq \langle 1, e\rangle] \\ \rightarrow \exists m \exists u(\langle m, u\rangle \in f \wedge \langle n, y\rangle = \langle s(m), k(u)\rangle))\end{aligned}$$

Let n and y be arbitrary such that $\langle n, y\rangle \in f$ and $\langle n, y\rangle \neq \langle 1, e\rangle$. Suppose, for contradiction, that the statement after the implication is false. Then $\forall m \forall u(\langle m, u\rangle \in f \rightarrow \langle n, y\rangle \neq \langle s(m), k(u)\rangle)$. The set $f - \{\langle n, y\rangle\}$ exists by the subset axiom; we will show that this set is a member of C. We know that $f \in C$, so $f \subseteq \mathbb{N} \times H$, and it is straightforward to show that $(f - \{\langle n, y\rangle\}) \subseteq \mathbb{N} \times H$. Since $\langle 1, e\rangle \neq \langle n, y\rangle$ and $\langle 1, e\rangle \in f$, we also have $\langle 1, e\rangle \in f - \{\langle n, y\rangle\}$. Finally, let arbitrary $\langle a, b\rangle \in f - \{\langle n, y\rangle\}$. Then $\langle a, b\rangle \in f \in C$ and $\langle s(a), k(b)\rangle \in f$. Since $\langle a, b\rangle \in f$, we must have, by our assumption, $\langle n, y\rangle \neq \langle s(a), k(b)\rangle$ and $\langle s(a), k(b)\rangle \in f - \{\langle n, y\rangle\}$. So $f - \{\langle n, y\rangle\} \in C$. It is straightforward to show that $f \not\subseteq f - \{\langle n, y\rangle\}$, contradicting the fact that $\forall X \in C(f \subseteq X)$.

We now prove functionality. We will do this by defining G:

$$\forall z(z \in G \leftrightarrow z \in \mathbb{N} \wedge \exists! x \in H(\langle z, x\rangle \in f))$$

(which exists by the subset axiom) and using the peano postulates to show that $G = \mathbb{N}$.
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We defined G:

$$\forall z(z \in G \leftrightarrow z \in \mathbb{N} \wedge \exists! x \in H(\langle z, x\rangle \in f))$$

To show that $G = \mathbb{N}$, we need to show that $G \subseteq \mathbb{N}$, $1 \in G$, and $\forall n \in G(s(n) \in G)$. The first requirement is straightforward. For the second, we already know by definition that $1 \in \mathbb{N}$. Since $f \in C$ we know by definition that $\langle 1, e\rangle \in f$, where $e \in H$. Let arbitrary $x' \in H$ such that $\langle 1, x'\rangle \in f$. Suppose for contradiction that $e \neq x'$. Then we have $\langle 1, e\rangle \neq \langle 1, x'\rangle$; so by the third statement proven earlier there exists m, u such that $\langle m, u\rangle \in f$ and $\langle 1, x'\rangle = \langle s(m), k(u)\rangle$. This means that $1 = s(m)$, where since $f \subseteq \mathbb{N} \times H$, $m \in \mathbb{N}$. This contradicts the peano postulates, which assert that $\neg\exists n \in \mathbb{N}(s(n) = 1)$.

For the third and final requirement, let n be an arbitrary element of G. Then by definition $n \in \mathbb{N}$ and there exists a unique x such that $\langle n, x\rangle \in f$. By definition we know $s(n) \in \mathbb{N}$, and since $f \in C$, we have $\langle s(n), k(x)\rangle \in f$, this verifies existence. Now let x' be arbitrary such that $\langle s(n), x'\rangle \in f$. By peano's postulates we know that $s(n) \neq 1$, so $\langle s(n), x'\rangle \neq \langle 1, e\rangle$. By the third statement proven earlier there exists m, u such that $\langle m, u\rangle \in f$ and $\langle s(n), x'\rangle = \langle s(m), k(u)\rangle$. So $s(n) = s(m)$ and $x' = k(u)$. Since s is injective we must have $m = n$. So $\langle n, u\rangle \in f$. By uniqueness of x we then have $u = x$. So $x' = k(u) = k(x)$, proving uniqueness. $s(n) \in G$.

We have shown that $G = \mathbb{N}$. It follows from this that $f : \mathbb{N} \to H$. For functionality, let $x \in \text{dom} f$. Then there exists y such that $\langle x, y\rangle \in f$. Since $f \subseteq \mathbb{N} \times H$ we have $x \in \mathbb{N} = G$, and by definition y is unique. Next to show $\text{dom} f = \mathbb{N}$, first let arbitrary $x \in \text{dom} f$, similarly it immediately follows that $x \in \mathbb{N}$. Now let $x \in \mathbb{N}$. Then $x \in G$ and by definition there exists $y \in H$ such that $\langle x, y\rangle \in f$, meaning $x \in \text{dom} f$. Finally to show $\text{ran} f \subseteq H$, let $y \in \text{ran} f$. Then there exists x such that $\langle x, y\rangle \in f \subseteq \mathbb{N} \times H$, so $y \in H$.

The fact that $\langle 1, e\rangle \in \bigcap C = f$ was already shown earlier. Finally we show $f \circ s = k \circ f$. Let $\langle x, y\rangle \in f \circ s$. Then $y = f \circ s(x) = f(s(x))$. Since $s(x) \neq 1$, $\langle s(x), y\rangle \neq \langle 1, e\rangle$ and as proven earlier there then exists m, u such that $\langle m, u\rangle \in f$ and $\langle s(x), y\rangle = \langle s(m), k(u)\rangle$. So $s(x) = s(m)$ and $y = k(u)$. Since s is injective $x = m$. We then have $\langle x, u\rangle \in f$ and $y = k(u) = k(f(x)) = k \circ f(x)$. so $\langle x, y\rangle \in k \circ f$. Next, suppose $\langle x, y\rangle \in k \circ f$. By definition there exists z such that $\langle x, z\rangle \in f$ and $\langle z, y\rangle \in k$. Since $f \in C$ we have $\langle s(x), k(z)\rangle \in f$. We then have $y = k(z) = f(s(x) = f \circ s(x)$ and $\langle x, y\rangle \in f \circ s$. We are done. $\square$

1.1.3 Addition on $\mathbb{N}$

Theorem 1.5. *There is a unique binary operation* $+ : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ *that satisfies the following two properties for all* $n, m \in \mathbb{N}$*:*

- $n + 1 = s(n)$
- $n + s(m) = s(n + m)$

Proof. We start with uniqueness. Let $+$ and $\oplus$ be arbitrary such that $+, \oplus : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$, $\forall m \in \mathbb{N} \forall n \in \mathbb{N}(+(\langle n, 1\rangle) = s(n) \wedge +(\langle n, s(m)\rangle) = s(+(\langle n, m\rangle)))$, and $\forall m \in \mathbb{N} \forall n \in \mathbb{N}(\oplus(\langle n, 1\rangle) = s(n) \wedge \oplus(\langle n, s(m)\rangle) = s(\oplus(\langle n, m\rangle)))$. We need to show $+ = \oplus$. We define G as

$$\forall x(x \in G \leftrightarrow x \in \mathbb{N} \wedge \forall n \in \mathbb{N}(+(\langle n, x\rangle) = \oplus(\langle n, x\rangle)))$$

(guaranteed by the subset axiom.) We will show that $G = \mathbb{N}$. To do this we need to show $G \subseteq \mathbb{N}$, $1 \in G$, and $\forall g \in G(s(g) \in G)$. The first requirement is straightforward. For the second requirement, we already know $1 \in \mathbb{N}$, and for arbitrary $n \in \mathbb{N}$, $+(\langle n, 1\rangle) = s(n) = \oplus(\langle n, 1\rangle)$.

For the third requirement, let arbitrary $g \in G$. Then by definition $g \in \mathbb{N}$ and $\forall n \in \mathbb{N}(+(\langle n, g\rangle) = \oplus(\langle n, g\rangle))$. Since $s : \mathbb{N} \to \mathbb{N}$ we know that $s(g) \in \mathbb{N}$ exists. Let n be arbitrary. By definition $+(\langle n, s(g)\rangle) = s(+(\langle n, g\rangle))$ and $\oplus(\langle n, s(g)\rangle) = s(\oplus(\langle n, g\rangle))$. By assumption we can assert that $+(\langle n, g\rangle) = \oplus(\langle n, g\rangle)$, so $+(\langle n, s(g)\rangle) = s(+(\langle n, g\rangle)) = s(\oplus(\langle n, g\rangle)) = \oplus(\langle n, s(g)\rangle)$. We conclude that $s(g) \in G$ and $G = \mathbb{N}$.

Let $\langle\langle x, y\rangle, z\rangle \in +$. By definition $y \in \mathbb{N} = G$ and $x \in \mathbb{N}$. So $+(\langle x, y\rangle) = \oplus(\langle x, y\rangle)$ and $z = \oplus(\langle x, y\rangle)$. So $\langle\langle x, y\rangle, z\rangle \in \oplus$. Now let $\langle\langle x, y\rangle, z\rangle \in \oplus$. A similar argument shows that $\langle\langle x, y\rangle, z\rangle \in +$.

Now for existence. We want to show

$$\begin{aligned}\exists + (+ : \mathbb{N} \times \mathbb{N} \to \mathbb{N} \wedge \forall m \in \mathbb{N} \forall n \in \mathbb{N}(+(\langle n, 1\rangle) = s(n) \\ \wedge +(\langle n, s(m)\rangle) = s(+(\langle n, m\rangle))))\end{aligned}$$

Theorem 1.4 allows us to assert

$$\forall k \forall H \forall e([e \in H \wedge k : H \to H] \to \exists! f(f : \mathbb{N} \to H \wedge f(1) = e \wedge f \circ s = k \circ f))$$

It immediately follows that

$$\forall p \in \mathbb{N} \exists! f(f : \mathbb{N} \to \mathbb{N} \wedge f(1) = p \wedge f \circ s = s \circ f)$$

We will define $+$ as the set where

$$\begin{aligned}\forall z(z \in + \leftrightarrow \exists a \in \mathbb{N} \exists b \in \mathbb{N}(\exists f(f : \mathbb{N} \to \mathbb{N} \wedge f(1) = s(a) \wedge f \circ s = s \circ f \\ \wedge \exists c(\langle b, c\rangle \in f \wedge z = \langle\langle a, b\rangle, c\rangle))))\end{aligned}$$

Existence follows from the subset axiom, where it can be shown that $+ \subseteq (\mathbb{N} \times \mathbb{N}) \times \mathbb{N}$.
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We defined $+$ as

$$\forall z(z \in + \leftrightarrow \exists a \in \mathbb{N} \exists b \in \mathbb{N}(\exists f(f : \mathbb{N} \to \mathbb{N} \wedge f(1) = s(a) \wedge f \circ s = s \circ f \\ \wedge \exists c(\langle b, c\rangle \in f \wedge z = \langle\langle a, b\rangle, c\rangle))))$$

First we prove functionality. Let x be an arbitrary element of $\mathrm{dom}(+)$. Then there exists y such that $\langle x, y\rangle \in +$. Let y' be arbitrary such that $\langle x, y'\rangle \in +$. $\langle x, y\rangle \in +$ by definition means that there exists $a, b \in \mathbb{N}$, and $f : \mathbb{N} \to \mathbb{N}$ such that $f(1) = s(a)$ and $f \circ s = s \circ f$, and that there exists c such that $\langle b, c\rangle \in f$ and $\langle x, y\rangle = \langle\langle a, b\rangle, c\rangle$. $\langle x, y'\rangle \in +$ similarly implies the existence of $a_0, b_0 \in \mathbb{N}$ and $f_0 : \mathbb{N} \to \mathbb{N}$ such that $f_0(1) = s(a_0)$ and $f_0 \circ s = s \circ f_0$, and the existence of c_0 such that $\langle b_0, c_0\rangle \in f_0$ and $\langle x, y'\rangle = \langle\langle a_0, b_0\rangle, c_0\rangle$. Then $\langle a, b\rangle = x = \langle a_0, b_0\rangle$, so $a_0 = a$ and $b_0 = b$. It follows that $s(a_0) = s(a) \in \mathbb{N}$. This and

$$\forall p \in \mathbb{N} \exists! f(f : \mathbb{N} \to \mathbb{N} \wedge f(1) = p \wedge f \circ s = s \circ f)$$

implies that $f = f_0$. So we have $y = c = f(b) = f(b_0) = f_0(b_0) = c_0 = y'$.

Next we show that $\mathrm{dom}(+) = \mathbb{N} \times \mathbb{N}$. First let x be an arbitrary element of $\mathrm{dom}(+)$. Then there exists y such that $\langle x, y\rangle \in +$. By definition there then exists $a, b \in \mathbb{N}$ and there exists c such that $\langle x, y\rangle = \langle\langle a, b\rangle, c\rangle$. So $x = \langle a, b\rangle \in \mathbb{N} \times \mathbb{N}$.

Now suppose $x \in \mathbb{N} \times \mathbb{N}$. Then by definition there exists $a, b \in \mathbb{N}$ such that $x = \langle a, b\rangle$. $s(a) \in \mathbb{N}$ exists, so by theorem 1.4 there exists f such that $f : \mathbb{N} \to \mathbb{N}$, $f(1) = s(a)$, and $f \circ s = s \circ f$. Since $b \in \mathbb{N} = \mathrm{dom} f$, $f(b)$ exists and $\langle b, f(b)\rangle \in f$. Then by definition, $\langle\langle a, b\rangle, f(b)\rangle \in +$, so $x = \langle a, b\rangle \in \mathrm{dom}(+)$.

Next we show that $\mathrm{ran}(+) \subseteq \mathbb{N}$. Let arbitrary $y \in \mathrm{ran}(+)$. Then there exists x such that $\langle x, y\rangle \in +$. By definition there then exists $a, b \in \mathbb{N}$, $f : \mathbb{N} \to \mathbb{N}$, and c such that $\langle b, c\rangle \in f$ and $\langle x, y\rangle = \langle\langle a, b\rangle, c\rangle$. So we have $y = c = f(b) \in \mathbb{N}$.

Now we verify the two properties of addition, that is,

$$\forall m \in \mathbb{N} \forall n \in \mathbb{N}(+(\langle n, 1\rangle) = s(n) \wedge +(\langle n, s(m)\rangle) = s(+(\langle n, m\rangle)))$$

Let arbitrary $n, m \in \mathbb{N}$. We know that $+ : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$, so $\langle n, 1\rangle \in \mathbb{N} \times \mathbb{N} = \mathrm{dom}(+)$ and there exists y such that $\langle\langle n, 1\rangle, y\rangle \in +$. By definition there then exists $a, b \in \mathbb{N}$ and $f : \mathbb{N} \to \mathbb{N}$ such that $f(1) = s(a)$, and there exists c such that $\langle b, c\rangle \in f$ and $\langle\langle n, 1\rangle, y\rangle = \langle\langle a, b\rangle, c\rangle$. Then $n = a$, $1 = b$, and $y = c$. We then have $\langle 1, y\rangle \in f$ and $y = f(1) = s(a) = s(n)$. We conclude that $\langle\langle n, 1\rangle, s(n)\rangle \in +$ and $+(\langle n, 1\rangle) = s(n)$.
(next page)

Now for the second property. We know that $s(m) \in \mathbb{N}$. So $\langle n, s(m)\rangle \in \mathbb{N}\times\mathbb{N} = \text{dom}(+)$ and there exists y such that $\langle\langle n, s(m)\rangle, y\rangle \in +$. By definition there then exists $a, b \in \mathbb{N}$ such that $f : \mathbb{N} \to \mathbb{N}$, $f(1) = s(a)$, and $f \circ s = s \circ f$, and there exists c such that $\langle b, c\rangle \in f$ and $\langle\langle n, s(m)\rangle, y\rangle = \langle\langle a, b\rangle, c\rangle$. We then have $n = a$, $s(m) = b$, and $y = c = f(b) = f(s(m)) = s(f(m))$. We know that $\langle n, m\rangle \in \mathbb{N}\times\mathbb{N} = \text{dom}(+)$, so there exists y_0 such that $\langle\langle n, m\rangle, y_0\rangle \in +$. Again by definition there exists $a_0, b_0 \in \mathbb{N}$ and $f_0 : \mathbb{N} \to \mathbb{N}$ such that $f_0(1) = s(a_0)$ and $f_0 \circ s = s \circ f_0$ and there exists c_0 such that $\langle b_0, c_0\rangle \in f_0$ and $\langle\langle n, m\rangle, y_0\rangle = \langle\langle a_0, b_0\rangle, c_0\rangle$. Since it follows that $a_0 = n = a$ and $s(a_0) = s(a)$, theorem 1.4 allows us to assert $f = f_0$. Putting it all together, we have $+(\langle n, m\rangle) = y_0 = c_0 = f_0(b_0) = f_0(m) = f(m)$. We conclude that $+(\langle n, s(m)\rangle) = y = s(f(m)) = s(+(\langle n, m\rangle))$. □

1.1.4 Multiplication on $\mathbb{N}$

Theorem 1.6. *There is a unique binary operation* $\cdot : \mathbb{N}\times\mathbb{N} \to \mathbb{N}$ *that satisfies the following two properties for all* $n, m \in \mathbb{N}$*:*

- $n \cdot 1 = n$
- $n \cdot s(m) = (n \cdot m) + n$

Proof. We start with uniqueness. Let $\cdot, \odot$ be arbitrary. Suppose $\cdot : \mathbb{N}\times\mathbb{N} \to \mathbb{N}$ and $\forall m \in \mathbb{N}\forall n \in \mathbb{N}(\cdot(\langle n, 1\rangle) = n \wedge \cdot(\langle n, s(m)\rangle) = +(\langle\cdot(\langle n, m\rangle), n\rangle))$. Similarly suppose $\odot : \mathbb{N}\times\mathbb{N} \to \mathbb{N}$ and $\forall m \in \mathbb{N}\forall n \in \mathbb{N}(\odot(\langle n, 1\rangle) = n \wedge \odot(\langle n, s(m)\rangle) = +(\langle\odot(\langle n, m\rangle), n\rangle))$. We want to show that $\cdot = \odot$. We do this by defining G such that

$$\forall z(z \in G \leftrightarrow z \in \mathbb{N} \wedge \forall n \in \mathbb{N}(\cdot(\langle n, z\rangle) = \odot(\langle n, z\rangle)))$$

(where existence is assured by the subset axiom) and showing that $G = \mathbb{N}$.

We need to show $G \subseteq \mathbb{N}$, $1 \in G$, and $\forall g \in G(s(g) \in G)$. The first requirement is straightforward. For the second, we already know $1 \in \mathbb{N}$, and see that for arbitrary $n \in \mathbb{N}$ we have $\cdot(\langle n, 1\rangle) = n = \odot(\langle n, 1\rangle)$.

For the third requirement let arbitrary $g \in G$. Then by definition $g \in \mathbb{N}$ and $\forall n \in \mathbb{N}(\cdot(\langle n, g\rangle) = \odot(\langle n, g\rangle))$. We know that $s(g) \in \mathbb{N}$ exists. So since $\langle n, s(g)\rangle \in \mathbb{N}\times\mathbb{N} = \text{dom}(\cdot)$, we have, by our assumptions, $\cdot(\langle n, s(g)\rangle) = +(\langle\cdot(\langle n, g\rangle), n\rangle) = +(\langle\odot(\langle n, g\rangle), n\rangle) = \odot(\langle n, s(g)\rangle)$. So $s(g) \in G$ and $G = \mathbb{N}$.

To prove the desired result, let arbitrary $z \in \cdot \subseteq (\mathbb{N}\times\mathbb{N})\times\mathbb{N}$. Then there exists $a_0, a_1, b \in \mathbb{N}$ such that $\langle\langle a_0, a_1\rangle, b\rangle = z \in \cdot$. So $b = \cdot(\langle a_0, a_1\rangle)$. Since $a_1 \in \mathbb{N} = G$, we can then conclude that $b = \cdot(\langle a_0, a_1\rangle) = \odot(\langle a_0, a_1\rangle)$. So $z = \langle\langle a_0, a_1\rangle, b\rangle \in \odot$. Next suppose $z \in \odot$. It then follows from a similar argument that $z \in \cdot$. We are done.
(next page)

Now for existence. We start by defining a function h for all q, where

$$\forall q \exists h \forall z(z \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N}(z = \langle m, x\rangle \wedge \langle\langle m, q\rangle, x\rangle \in +))$$

(which exists by the subset axiom, where $\mathbb{N} \times \mathbb{N}$ is a bounding set. Uniqueness for each q follows from extensionality.) We show that for all $q \in \mathbb{N}$, this h is a function from $\mathbb{N}$ into $\mathbb{N}$. That is

$$\forall q \in \mathbb{N}[\forall z(z \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N}(z = \langle m, x\rangle \wedge \langle\langle m, q\rangle, x\rangle \in +)) \rightarrow h : \mathbb{N} \rightarrow \mathbb{N}]$$

Let arbitrary $q \in \mathbb{N}$. Suppose that the statement before the implication is true. First to prove functionality. Let $a \in \mathrm{dom} h$. Then there exists b such that $\langle a, b\rangle \in h$. Let b' be arbitrary such that $\langle a, b'\rangle \in h$. By our assumptions we must have $a, b, b' \in \mathbb{N}$, $+(\langle a, q\rangle) = b$, and $+(\langle a, q\rangle) = b'$. Since $+ : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ we must have $b = b'$. Next for the domain bound suppose arbitrary $a \in \mathrm{dom} h$. Then there exists b such that $\langle a, b\rangle \in h$. It immediately follows that $a \in \mathbb{N}$. Suppose instead that $a \in \mathbb{N}$. Then $\langle a, q\rangle \in \mathbb{N} \times \mathbb{N} = \mathrm{dom}(+)$. So $+(\langle a, q\rangle) \in \mathbb{N}$ exists and by definition, $\langle a, +(\langle a, q\rangle)\rangle \in h$. So $a \in \mathrm{dom} h$. Finally for the range bound, let arbitrary $b \in \mathrm{ran} h$. Then there exists a such that $\langle a, b\rangle \in h$. It immediately follows by definition that $b \in \mathbb{N}$. We have therefore shown

$$\begin{gathered}\forall q \in \mathbb{N} \exists! h(\forall z(z \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N}(z = \langle m, x\rangle \wedge \langle\langle m, q\rangle, x\rangle \in +)) \\ \wedge\, h : \mathbb{N} \rightarrow \mathbb{N})\end{gathered}$$

Theorem 1.4 states

$$\forall H \forall q \forall h([q \in H \wedge h : H \rightarrow H] \rightarrow \exists! g(g : \mathbb{N} \rightarrow H \wedge g(1) = q \wedge g \circ s = h \circ g))$$

which allows us to assert

$$\forall q \forall h([q \in \mathbb{N} \wedge h : \mathbb{N} \rightarrow \mathbb{N}] \rightarrow \exists! g(g : \mathbb{N} \rightarrow \mathbb{N} \wedge g(1) = q \wedge g \circ s = h \circ g))$$

(next page)

We showed

$$\forall q \in \mathbb{N} \exists! h(\forall z(z \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N}(z = \langle m, x\rangle \wedge \langle\langle m, q\rangle, x\rangle \in +)) \\ \wedge\, h : \mathbb{N} \to \mathbb{N})$$

and

$$\forall q \forall h([q \in \mathbb{N} \wedge h : \mathbb{N} \to \mathbb{N}] \to \exists! g(g : \mathbb{N} \to \mathbb{N} \wedge g(1) = q \wedge g \circ s = h \circ g))$$

Now we construct $\cdot$ as the set

$$\begin{aligned} &\forall z(z \in \cdot \leftrightarrow \exists a \in \mathbb{N} \exists b \in \mathbb{N}(\\ &\qquad \exists h(\forall z_0(z_0 \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N}(z_0 = \langle m, x\rangle \wedge \langle\langle m, a\rangle, x\rangle \in +)) \\ &\qquad\quad \wedge\, h : \mathbb{N} \to \mathbb{N} \wedge \exists g(g : \mathbb{N} \to \mathbb{N} \wedge g(1) = a \wedge g \circ s = h \circ g \\ &\qquad\quad \wedge\, \exists c(\langle b, c\rangle \in g \wedge z = \langle\langle a, b\rangle, c\rangle)))))) \end{aligned}$$

(existence follows from the subset axiom, where $(\mathbb{N} \times \mathbb{N}) \times \mathbb{N}$ is a bounding set.) We want to verify

$$\cdot : \mathbb{N} \times \mathbb{N} \to \mathbb{N} \wedge \forall m \in \mathbb{N} \forall n \in \mathbb{N}(\cdot(\langle n, 1\rangle) = n \wedge \cdot(\langle n, s(m)\rangle) = +(\langle \cdot(\langle n, m\rangle), n\rangle))$$

We start with functionality. Let arbitrary $p \in \mathrm{dom}(\cdot)$. Then by definition there exists q such that $\langle p, q\rangle \in \cdot$. This means that there exists $a, b \in \mathbb{N}$ and $h : \mathbb{N} \to \mathbb{N}$ such that $\forall z_0(z_0 \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N}(z_0 = \langle m, x\rangle \wedge \langle\langle m, a\rangle, x\rangle \in +))$. There also exists $g : \mathbb{N} \to \mathbb{N}$ such that $g(1) = a$ and $g \circ s = h \circ g$, and c such that $\langle b, c\rangle \in g$ and $\langle p, q\rangle = \langle\langle a, b\rangle, c\rangle$. Now suppose there exists $\langle p, q'\rangle \in \cdot$. Similarly there exists $a', b' \in \mathbb{N}$, $h' : \mathbb{N} \to \mathbb{N}$ such that $\forall z_0(z_0 \in h' \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N}(z_0 = \langle m, x\rangle \wedge \langle\langle m, a'\rangle, x\rangle \in +))$, $g' : \mathbb{N} \to \mathbb{N}$ such that $g'(1) = a'$ and $g' \circ s = h' \circ g'$, and c' such that $\langle b', c'\rangle \in g'$ and $\langle p, q'\rangle = \langle\langle a', b'\rangle, c'\rangle$. We have $\langle a', b'\rangle = p = \langle a, b\rangle$. So $a' = a$ and $b' = b$. Uniqueness, as proven, implies that $h = h'$ and $g = g'$. So $q' = c' = g'(b') = g(b') = g(b) = c = q$.

Next we show $\mathrm{dom}(\cdot) = \mathbb{N} \times \mathbb{N}$. Let arbitrary $p \in \mathrm{dom}(\cdot)$. Then by definition there exists q such that $\langle p, q\rangle \in \cdot$. So there exists $a, b \in \mathbb{N}$ and c such that $\langle p, q\rangle = \langle\langle a, b\rangle, c\rangle$. So $p = \langle a, b\rangle \in \mathbb{N} \times \mathbb{N}$. Now let $p \in \mathbb{N} \times \mathbb{N}$. Then by definition there exists $a, b \in \mathbb{N}$ such that $p = \langle a, b\rangle$. Since $a \in \mathbb{N}$, as proven there exists $h : \mathbb{N} \to \mathbb{N}$ such that $\forall z(z \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N}(z = \langle m, x\rangle \wedge \langle\langle m, a\rangle, x\rangle \in +))$, and there exists $g : \mathbb{N} \to \mathbb{N}$ such that $g(1) = a$ and $g \circ s = h \circ g$. By definition we then have $\langle\langle a, b\rangle, g(b)\rangle \in \cdot$ and $p = \langle a, b\rangle \in \mathrm{dom}(\cdot)$.

For the range bound, let $q \in \mathrm{ran}(\cdot)$. Then there exists p such that $\langle p, q\rangle \in \cdot$. So there exists $a, b \in \mathbb{N}$, $g : \mathbb{N} \to \mathbb{N}$, and c such that $\langle b, c\rangle \in g$ and $\langle p, q\rangle = \langle\langle a, b\rangle, c\rangle$. So $q = c = g(b) \in \mathbb{N}$.

Finally we verify the two properties of multiplication. Let arbitrary $m, n \in \mathbb{N}$. We first show $\cdot(\langle n, 1\rangle) = n$. We know that $\langle\langle n, 1\rangle, \cdot(\langle n, 1\rangle)\rangle \in \cdot$. By definition there then exists $a, b \in \mathbb{N}$, $g : \mathbb{N} \to \mathbb{N}$ such that $g(1) = a$, and c such that $\langle b, c\rangle \in g$ and $\langle\langle n, 1\rangle, \cdot(\langle n, 1\rangle)\rangle = \langle\langle a, b\rangle, c\rangle$. So $\cdot(\langle n, 1\rangle) = c = g(b) = g(1) = a = n$. (next page)

Now we show $\cdot(\langle n, s(m)\rangle) = +(\langle\cdot(\langle n, m\rangle), n\rangle)$. $s(m) \in \mathbb{N}$ exists, so we have $\langle\langle n, s(m)\rangle, \cdot(\langle n, s(m)\rangle)\rangle \in \cdot$. Then there exists $a, b \in \mathbb{N}$, $h : \mathbb{N} \to \mathbb{N}$ such that $\forall z_0(z_0 \in h \leftrightarrow \exists m_0 \in \mathbb{N}\exists x \in \mathbb{N}(z_0 = \langle m_0, x\rangle \wedge \langle\langle m_0, a\rangle, x\rangle \in +))$, $g : \mathbb{N} \to \mathbb{N}$ such that $g(1) = a$ and $g \circ s = h \circ g$, and c such that $\langle b, c\rangle \in g$ and $\langle\langle n, s(m)\rangle, \cdot(\langle n, s(m)\rangle)\rangle = \langle\langle a, b\rangle, c\rangle$. We then have $\cdot(\langle n, s(m)\rangle) = c = g(b) = g(s(m)) = h(g(m))$ and $a = n$. Similarly we know that $\langle\langle n, m\rangle, \cdot(\langle n, m\rangle)\rangle \in \cdot$. So there exists $a_0, b_0 \in \mathbb{N}$, $h_0 : \mathbb{N} \to \mathbb{N}$ such that $\forall z_0(z_0 \in h_0 \leftrightarrow \exists m_0 \in \mathbb{N}\exists x \in \mathbb{N}(z_0 = \langle m_0, x\rangle \wedge \langle\langle m_0, a_0\rangle, x\rangle \in +))$, $g_0 : \mathbb{N} \to \mathbb{N}$ such that $g_0(1) = a_0$ and $g_0 \circ s = h_0 \circ g_0$, and c_0 such that $\langle b_0, c_0\rangle \in g_0$ and $\langle\langle n, m\rangle, \cdot(\langle n, m\rangle)\rangle = \langle\langle a_0, b_0\rangle, c_0\rangle$. This means that $a_0 = n = a$, and that by our earlier statements we can assert that $h = h_0$, and $g = g_0$. So $\cdot(\langle n, m\rangle) = c_0 = g_0(b_0) = g(m)$. We then have $\cdot(\langle n, s(m)\rangle) = h(g(m)) = h(\cdot(\langle n, m\rangle))$. From the definition of h, there then exists $m_0, x \in \mathbb{N}$ such that $\langle\cdot(\langle n, m\rangle), h(\cdot(\langle n, m\rangle))\rangle = \langle m_0, x\rangle$ and $\langle\langle m_0, a\rangle, x\rangle \in +$. This and the fact that $a = n$ allows us to conclude $+(\langle\cdot(\langle n, m\rangle), n\rangle) = h(\cdot(\langle n, m\rangle)) = \cdot(\langle n, s(m)\rangle)$. $\square$

1.1.5 Properties of addition and multiplication on the naturals

Theorem 1.7. *Let $a, b, c \in \mathbb{N}$.*

1. *If $a + c = b + c$, then $a = b$ (Cancellation Law for Addition)*
2. *$(a + b) + c = a + (b + c)$ (Associative Law for Addition)*
3. *$1 + a = s(a) = a + 1$*
4. *$a + b = b + a$ (Commutative Law for Addition)*
5. *$a + b \neq 1$*
6. *$a + b \neq a$*
7. *$a \cdot 1 = a = 1 \cdot a$ (Identity Law for Multiplication)*
8. *$(a + b)c = ac + bc$ (Distributive Law)*
9. *$ab = ba$ (Commutative Law for Multiplication)*
10. *$c(a + b) = ca + cb$ (Distributive Law)*
11. *$(ab)c = a(bc)$ (Associative Law for Multiplication)*
12. *If $ac = bc$ then $a = b$ (Cancellation Law for Multiplication)*
13. *$ab = 1$ iff $a = 1 = b$*

(next page)

Proof. (1) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N}(a + c = b + c \to a = b)$$

To do this we define the set G as the set where

$$\forall c(c \in G \leftrightarrow c \in \mathbb{N} \land \forall a \in \mathbb{N} \forall b \in \mathbb{N}(a + c = b + c \to a = b))$$

(where existence follows from the subset axiom using $\mathbb{N}$ as the bounding set. Uniqueness follows from extensionality.) Then we show that $G = \mathbb{N}$. Showing $G \subseteq \mathbb{N}$ is straightforward; we need to show $1 \in G$ and $\forall g \in G(s(g) \in G)$. For the former, we already know $1 \in \mathbb{N}$. Let arbitrary $a, b \in \mathbb{N}$. Suppose that $a + 1 = b + 1$. By definition we then have $s(a) = s(b)$, so by injectivity $a = b$ and $1 \in G$.

Now the latter. Let arbitrary $g \in G$. As defined we then have $g \in \mathbb{N}$ and $\forall a \in \mathbb{N} \forall b \in \mathbb{N}(a + g = b + g \to a = b)$. $s(g) \in \mathbb{N}$ exists. Let arbitrary $a, b \in \mathbb{N}$ and suppose that $a + s(g) = b + s(g)$. Then by the definition of addition we have $s(a + g) = s(b + g)$ and, by injectivity, $a + g = b + g$. Since $g \in G$ we then have $a = b$. So $G = \mathbb{N}$. To prove the initial statement, let arbitrary $c \in \mathbb{N}$. Then from $\mathbb{N} = G$ the desired conclusion immediately follows. □

Proof. (2) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N}((a + b) + c = a + (b + c))$$

To do this we define G as the set where

$$\forall c(c \in G \leftrightarrow c \in \mathbb{N} \land \forall a \in \mathbb{N} \forall b \in \mathbb{N}((a + b) + c = a + (b + c)))$$

(where existence and uniqueness can be easily shown) and show that $G = \mathbb{N}$. Showing $G \subseteq \mathbb{N}$ is straightforward. We need to show $1 \in G$ and $\forall g \in G(s(g) \in G)$.

For the former, we already know $1 \in \mathbb{N}$. Let arbitrary $a, b \in \mathbb{N}$. By the definition of addition we have

$$(a + b) + 1 = s(a + b) = a + s(b) = a + (b + 1)$$

Now the latter. Let arbitrary $g \in G$. Then by definition $g \in \mathbb{N}$ and $\forall a \in \mathbb{N} \forall b \in \mathbb{N}((a + b) + g = a + (b + g))$. We know $s(g) \in \mathbb{N}$ exists. Let arbitrary $a, b \in \mathbb{N}$. By definition we then have

$$(a + b) + s(g) = s((a + b) + g) = s(a + (b + g)) = a + s(b + g) = a + (b + s(g))$$

So $\mathbb{N} = G$. To prove the initial statement, let arbitrary $c \in \mathbb{N} = G$. The desired conclusion immediately follows. □

(next page)

Proof. (3) We want to show

$$\forall a \in \mathbb{N}(1 + a = s(a) = a + 1)$$

To do this we define G as the set where

$$\forall a(a \in G \leftrightarrow a \in \mathbb{N} \wedge 1 + a = s(a) = a + 1)$$

(Showing existence and uniqueness is straightforward.) We show that $G = \mathbb{N}$. Showing $G \subseteq \mathbb{N}$ is straightforward. We need to show that $1 \in G$ and $\forall g \in G(s(g) \in G)$. For the former we already know that $1 \in \mathbb{N}$, and by definition we have $1 + 1 = s(1)$.

Now the latter. Let arbitrary $g \in G$. Then by definition $g \in \mathbb{N}$ and $1 + g = s(g) = g + 1$. We know $s(g) \in \mathbb{N}$ exists. We then have

$$1 + s(g) = s(1 + g) = s(s(g)) = s(g + 1) = (g + 1) + 1 = s(g) + 1$$

So $G = \mathbb{N}$. Proving the initial statement is straightforward. Let arbitrary $a \in \mathbb{N} = G$ and the desired conclusion immediately follows. $\square$

Proof. (4) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(a + b = b + a)$$

To do this we define G as the set such that

$$\forall a(a \in G \leftrightarrow a \in \mathbb{N} \wedge \forall b \in \mathbb{N}(a + b = b + a))$$

(Showing existence and uniqueness is straightforward.) We show that $G = \mathbb{N}$. Showing $G \subseteq \mathbb{N}$ is straightforward. We need to show that $1 \in G$ and $\forall g \in G(s(g) \in G)$. For the former we already know $1 \in \mathbb{N}$. Letting arbitrary $b \in \mathbb{N}$, by (3) we have $1 + b = b + 1$.

Now the latter. Let arbitrary $g \in G$. Then by definition $g \in \mathbb{N}$ and $\forall b \in \mathbb{N}(g + b = b + g)$. $s(g) \in \mathbb{N}$ exists. Let arbitrary $b \in \mathbb{N}$. Then we have

$$s(g) + b = (g + 1) + b \overset{(2)}{=} g + (1 + b) \overset{(3)}{=} g + s(b) = s(g + b) = s(b + g) = b + s(g)$$

So $G = \mathbb{N}$. Showing the initial statement is straightforward. Let arbitrary $a \in \mathbb{N} = G$ and the desired conclusion immediately follows. $\square$

(next page)

Proof. (5) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(a + b \neq 1)$$

Let arbitrary $a, b \in \mathbb{N}$. Suppose for contradiction that $a + b = 1$. We have 2 cases. First suppose $b = 1$. Then $a + b = a + 1 = s(a) = 1$, and the last equality is a contradiction of the peano postulates. Now suppose $b \neq 1$. Then by lemma 1.3 there exists $x \in \mathbb{N}$ such that $b = s(x)$. We then have $a + b = a + s(x) = s(a + x) = 1$, which is again a contradiction. □

Proof. (6) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(a + b \neq a)$$

We do this by defining G as the set such that

$$\forall a(a \in G \leftrightarrow a \in \mathbb{N} \wedge \forall b \in \mathbb{N}(a + b \neq a))$$

(Showing existence and uniqueness is straightforward.) We show that $G = \mathbb{N}$. Showing $G \subseteq \mathbb{N}$ is straightforward. We need to show $1 \in G$ and $\forall g \in G(s(g) \in G)$. For the former, we already know $1 \in \mathbb{N}$. Let arbitrary $b \in \mathbb{N}$. By (5) $1 + b \neq 1$.

Now the latter. Let arbitrary $g \in G$. Then $g \in \mathbb{N}$ and $\forall b \in \mathbb{N}(g + b \neq g)$. $s(g) \in \mathbb{N}$ exists. Let arbitrary $b \in \mathbb{N}$. For contradiction, suppose $s(g) + b = s(g)$. We then have

$$s(g) = s(g) + b \stackrel{(4)}{=} b + s(g) = s(b + g)$$

By injectivity of s and (4) we then have $g = b + g = g + b$. This contradicts $\forall b \in \mathbb{N}(g + b \neq g)$. So $G = \mathbb{N}$. Showing the initial statement is straightforward. Let arbitrary $a \in \mathbb{N} = G$ and the desired conclusion immediately follows. □

Proof. (7) We want to show

$$\forall a \in \mathbb{N}(a \cdot 1 = a = 1 \cdot a)$$

to do this we define G as the set such that

$$\forall a(a \in G \leftrightarrow a \in \mathbb{N} \wedge a \cdot 1 = a = 1 \cdot a)$$

(where existence and uniqueness can be easily shown.) We show that $G = \mathbb{N}$. Showing $G \subseteq \mathbb{N}$ is straightforward. We need to show $1 \in G$ and $\forall g \in G(s(g) \in G)$. For the former we already know $1 \in \mathbb{N}$, and $1 \cdot 1 = 1$.

Now the latter. Let arbitrary $g \in G$. Then $g \in \mathbb{N}$ and $g \cdot 1 = g = 1 \cdot g$. $s(g) \in \mathbb{N}$ exists. We then have

$$s(g) \cdot 1 = s(g) = s(g \cdot 1) = s(1 \cdot g) = (1 \cdot g) + 1 = 1 \cdot s(g)$$

so $G = \mathbb{N}$. Proving the initial statement is straightforward. Let arbitrary $a \in \mathbb{N} = G$ and the desired conclusion immediately follows. □

(next page)

Proof. (8) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N}((a+b)c = (ac) + (bc))$$

To do this we define G as the set such that

$$\forall c(c \in G \leftrightarrow c \in \mathbb{N} \land \forall a \in \mathbb{N} \forall b \in \mathbb{N}((a+b)c = (ac) + (bc))$$

(where proving existence and uniqueness is straightforward.) We show that $G = \mathbb{N}$. Showing $G \subseteq \mathbb{N}$ is straightforward. We need to show $1 \in G$ and $\forall g \in G(s(g) \in G)$. For the former, we already know $1 \in \mathbb{N}$. Let arbitrary $a, b \in \mathbb{N}$. We have

$$(a+b) \cdot 1 = a + b = (a \cdot 1) + (b \cdot 1)$$

Now the latter. Let arbitrary $g \in G$. Then by definition $g \in \mathbb{N}$ and $\forall a \in \mathbb{N} \forall b \in \mathbb{N}((a+b)g = (ag) + (bg))$. $s(g) \in \mathbb{N}$ exists. Let arbitrary $a, b \in \mathbb{N}$. We have

$$\begin{aligned}
(a+b)s(g) &= ((a+b)g) + (a+b) = ((ag) + (bg)) + (a+b) \\
&\overset{(2)}{=} (ag) + ((bg) + (a+b)) \overset{(4,2)}{=} (ag) + ((bg+b) + a) \\
&= (ag) + ((bs(g)) + a) \overset{(4,2)}{=} ((ag) + a) + (bs(g)) \\
&= (as(g)) + (bs(g))
\end{aligned}$$

So $G = \mathbb{N}$. Proving the initial statement is straightforward. Let arbitrary $c \in \mathbb{N} = G$ and the desired conclusion immediately follows. □

Proof. (9) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(ab = ba)$$

To do this we define G as the set such that

$$\forall a(a \in G \leftrightarrow a \in \mathbb{N} \land \forall b \in \mathbb{N}(ab = ba))$$

(where proving existence and uniqueness is straightforward.) We show that $G = \mathbb{N}$. Showing $G \subseteq \mathbb{N}$ is straightforward. We need to show $1 \in G$ and $\forall g \in G(s(g) \in G)$. For the former, we know $1 \in \mathbb{N}$. Let arbitrary $b \in \mathbb{N}$. Then by (7) we have $1 \cdot b = b \cdot 1$.

Now the latter. Let arbitrary $g \in G$. Then $g \in \mathbb{N}$ and $\forall b \in \mathbb{N}(gb = bg)$. $s(g) \in \mathbb{N}$ exists. Let arbitrary $b \in \mathbb{N}$. We have

$$s(g)b = (g+1)b \overset{(8)}{=} (gb) + (1 \cdot b) \overset{(7)}{=} (bg) + b = bs(g)$$

So $G = \mathbb{N}$. Proving the initial statement is straightforward. Let arbitrary $a \in \mathbb{N} = G$ and the desired conclusion immediately follows. □

(next page)

Proof. (10) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N}(c(a+b) = (ca) + (cb))$$

Let arbitrary $a, b, c \in \mathbb{N}$. We have

$$c(a+b) \overset{(9)}{=} (a+b)c \overset{(8)}{=} (ac) + (bc) \overset{(9)}{=} (ca) + (cb)$$ □

Proof. (11) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N}((ab)c = a(bc))$$

To do this we define G as the set such that

$$\forall b(b \in G \leftrightarrow b \in \mathbb{N} \wedge \forall a \in \mathbb{N} \forall c \in \mathbb{N}((ab)c = a(bc)))$$

(where proving existence and uniqueness is straightforward.) We show that $G = \mathbb{N}$. Showing $G \subseteq \mathbb{N}$ is straightforward. We need to show $1 \in G$ and $\forall g \in G(s(g) \in G)$. For the former, we already know $1 \in \mathbb{N}$. Let arbitrary $a, c \in \mathbb{N}$. We have

$$(a \cdot 1)c = ac \overset{(7)}{=} a(1 \cdot c)$$

Now the latter. Let arbitrary $g \in G$. Then $g \in \mathbb{N}$ and $\forall a \in \mathbb{N} \forall c \in \mathbb{N}((ag)c = a(gc))$. $s(g) \in \mathbb{N}$ exists. Let arbitrary $a, c \in \mathbb{N}$. We have

$$\begin{aligned} (as(g))c &= ((ag) + a)c \overset{(8)}{=} ((ag)c) + (ac) = (a(gc)) + (ac) \\ &\overset{(10)}{=} a((gc) + c) \overset{(9)}{=} a((cg) + c) = a(cs(g)) \\ &\overset{(9)}{=} a(s(g)c) \end{aligned}$$

So $G = \mathbb{N}$. Proving the initial statement is straightforward. Let arbitrary $b \in \mathbb{N} = G$ and the desired conclusion immediately follows. □

(next page)

Proof. (12) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N}(ac = bc \to a = b)$$

To do this we define G as the set such that

$$\forall a(a \in G \leftrightarrow a \in \mathbb{N} \land \forall b \in \mathbb{N} \forall c \in \mathbb{N}(ac = bc \to a = b))$$

(where proving existence and uniqueness is straightforward.) We show that $G = \mathbb{N}$. Showing $G \subseteq \mathbb{N}$ is straightforward. We need to show $1 \in G$ and $\forall g \in G(s(g) \in G)$.

For the former, we already know $1 \in \mathbb{N}$. Let arbitrary $b, c \in \mathbb{N}$ and suppose that $1 \cdot c = bc$. Suppose for contradiction that $b \neq 1$. Then by lemma 1.3 there exists $x \in \mathbb{N}$ such that $b = s(x)$. We then have

$$c \overset{(7)}{=} 1 \cdot c = bc = s(x)c = (x+1)c \overset{(8,7)}{=} (xc) + c \overset{(4,9)}{=} c + (cx)$$

which contradicts (6).

Now the latter. Let arbitrary $g \in G$. Then $g \in \mathbb{N}$ and $\forall b \in \mathbb{N} \forall c \in \mathbb{N}(gc = bc \to g = b)$. $s(g) \in \mathbb{N}$ exists. Let arbitrary $b, c \in \mathbb{N}$. Suppose $s(g)c = bc$. For contradiction, suppose $b = 1$. Then

$$c \overset{(7)}{=} 1 \cdot c = bc = s(g)c \overset{(9)}{=} cs(g) = (cg) + c \overset{(4)}{=} c + (cg)$$

contradicting (6). So we must have $b \neq 1$ and by lemma 1.3 there exists $x \in \mathbb{N}$ such that $b = s(x)$. Then

$$(cg) + c = cs(g) \overset{(9)}{=} s(g)c = bc = s(x)c = (x+1)c \overset{(8,7)}{=} (xc) + c$$

and by (1), $gc = cg = xc$. Since $g \in G$ we then have $g = x$ and $s(g) = s(x) = b$. So $G = \mathbb{N}$. Proving the initial statement is straightforward. Let arbitrary $a \in \mathbb{N} = G$ and the desired conclusion immediately follows. $\square$

Proof. (13) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(ab = 1 \leftrightarrow a = 1 = b)$$

Let arbitrary $a, b \in \mathbb{N}$. First suppose $ab = 1$. For contradiction suppose $b \neq 1$. Then by lemma 1.3 there exists $x \in \mathbb{N}$ such that $b = s(x)$. So

$$1 = ab = as(x) = a(x+1) \overset{(10)}{=} (ax) + (a \cdot 1) = (ax) + a$$

which contradicts (5). So $b = 1$. We immediately have $1 = ab = a \cdot 1 = a$ and $a = 1 = b$. Now suppose $a = 1 = b$. We immediately have $ab = 1 \cdot 1 = 1$. $\square$

1.1.6 Ordering on $\mathbb{N}$

Definition 1.8. *The relation $<$ on $\mathbb{N}$ is defined by $a < b$ if and only if there is some $p \in \mathbb{N}$ such that $a + p = b$ for all $a, b \in \mathbb{N}$.*

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(a < b \leftrightarrow \exists p \in \mathbb{N}(a + p = b))$$

The relation $\leq$ on $\mathbb{N}$ is defined by $a \leq b$ if and only if $a < b$ or $a = b$ for all $a, b \in \mathbb{N}$.

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(a \leq b \leftrightarrow (a < b \vee a = b))$$

We will write $a > b$ to mean the same thing as $b < a$. The same applies to $a \geq b$.

Theorem 1.9. *Let $a, b \in \mathbb{N}$. Suppose that $a < b$. Then there is a unique $p \in \mathbb{N}$ such that $a + p = b$.*

Proof. By definition there exists $p \in \mathbb{N}$ such that $a + p = b$. Let $p' \in \mathbb{N}$ be arbitrary such that $a + p' = b$. Then by part 4 of theorem 1.7 we have $p + a = b$ and $p' + a = b$, so $p + a = p' + a$. By part 1 of theorem 1.7 we have $p = p'$. □

Theorem 1.10. *Let $a, b, c \in \mathbb{N}$.*

1. *$a \leq a$, and $a \not< a$, and $a < a + 1$*
2. *$1 \leq a$*
3. *If $a < b$ and $b < c$, then $a < c$; if $a \leq b$ and $b < c$, then $a < c$; if $a < b$ and $b \leq c$, then $a < c$; if $a \leq b$ and $b \leq c$, then $a \leq c$*
4. *$a < b$ if and only if $a + c < b + c$*
5. *Precisely one of $a < b$ or $a = b$ or $a > b$ holds (Trichotomy Law)*
6. *$a < b$ if and only if $ac < bc$*
7. *$a \leq b$ or $b \leq a$*
8. *If $a \leq b$ and $b \leq a$, then $a = b$*
9. *It cannot be that $b < a < b + 1$*
10. *$a \leq b$ if and only if $a < b + 1$*
11. *$a < b$ if and only if $a + 1 \leq b$*

(next page)

Proof. (1) We want to show

$$\forall a \in \mathbb{N}(a \leq a \wedge a \not< a \wedge a < a+1)$$

Let arbitrary $a \in \mathbb{N}$. Since $a = a$, by definition $a \leq a$. Next suppose for contradiction that $a < a$. Then there would exist $p \in \mathbb{N}$ such that $a + p = a$, contradicting part 6 of theorem 1.7. Finally since $1 \in \mathbb{N}$ and $a + 1 = a + 1$, by definition $a < a + 1$. $\square$

Proof. (2) We want to show

$$\forall a \in \mathbb{N}(1 \leq a)$$

Let arbitrary $a \in \mathbb{N}$. We have 2 cases. First suppose $a = 1$; then by definition $1 \leq a$. Next suppose $a \neq 1$. Then by lemma 1.3 there exists $x \in \mathbb{N}$ such that $a = s(x) = x + 1$. By theorem 1.7 part 4 we have $a = 1 + x$. So by definition $1 < a$. $\square$

Proof. (3) We want to show

$$\begin{aligned} \forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N}([(a < b \wedge b < c) \to a < c] \wedge [(a \leq b \wedge b < c) \to a < c] \\ \wedge [(a < b \wedge b \leq c) \to a < c] \wedge [(a \leq b \wedge b \leq c) \to a \leq c]) \end{aligned}$$

Let arbitrary $a, b, c \in \mathbb{N}$. First suppose $a < b \wedge b < c$. Then there exists $x, y \in \mathbb{N}$ such that $a + x = b$ and $b + y = c$. We then have $(a + x) + y = b + y = c$. By theorem 1.7 part 2 we have $a + (x + y) = c$ and therefore $a < c$.

Next suppose $a \leq b \wedge b < c$. There exists $y \in \mathbb{N}$ such that $b + y = c$. We have 2 cases. First suppose $a < b$. Then as proven earlier we have $a < c$. Next suppose $a = b$. Then we immediately have $a + y = c$ and $a < c$.

Next suppose $a < b \wedge b \leq c$. There exists $x \in \mathbb{N}$ such that $a + x = b$. We have 2 cases. First suppose $b < c$. Then again we immediately have $a < c$. Next suppose $b = c$. Then $a + x = c$ and $a < c$.

Finally suppose $a \leq b \wedge b \leq c$. We have 4 cases. First suppose $a < b \wedge b < c$. As proven $a < c$. Next suppose $a = b \wedge b < c$. As proven $a < c$. Next suppose $a < b \wedge b = c$. As proven $a < c$. Finally suppose $a = b$ and $b = c$. It immediately follows that $a = c$ and $a \leq c$. $\square$

Proof. (4) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N}(a < b \leftrightarrow a + c < b + c)$$

Let arbitrary $a, b, c \in \mathbb{N}$. First suppose $a < b$. Then there exists $x \in \mathbb{N}$ such that $a + x = b$. Then $(a + x) + c = b + c$ and by theorem 1.7 parts 2 and 4 we have $(a + c) + x = b + c$. So $a + c < b + c$.

Next suppose $a + c < b + c$. Then there exists $x \in \mathbb{N}$ such that $(a + c) + x = b + c$. By theorem 1.7 parts 2 and 4 we have $(a + x) + c = b + c$, and by part 1 of that same theorem we have $a + x = b$. So $a < b$. $\square$

(next page)

Proof. (5) We want to prove trichotomy

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}((a < b \lor a = b \lor b < a) \\ \land \neg((a < b) \land (a = b)) \land \neg((a < b) \land (b < a)) \land \neg((a = b) \land (b < a)))$$

Let arbitrary $a, b \in \mathbb{N}$. We start by proving the last three statements. First suppose for contradiction that $a < b$ and $a = b$. Then $a < a$, contradicting (1). Next suppose for contradiction that $a = b$ and $b < a$. A similar argument shows that this contradicts (1). Finally suppose for contradiction that $a < b \land b < a$. By (3) we have $a < a$, a contradiction.

Now for the first statement. We define G as the set such that

$$\forall a(a \in G \leftrightarrow a \in \mathbb{N} \land \forall b \in \mathbb{N}(a < b \lor a = b \lor b < a))$$

(proving existence and uniqueness of G is straightforward.) We show that $G = \mathbb{N}$. Showing $G \subseteq \mathbb{N}$ is straightforward. We need to show that $1 \in G$ and $\forall g \in G(s(g) \in G)$. For the former, we already know $1 \in \mathbb{N}$. Let arbitrary $b \in \mathbb{N}$. By (2) we have $1 \leq b$, meaning $1 < b$ or $1 = b$. Either case allows us to conclude that $1 \in G$.

Now the latter. Let arbitrary $g \in G$. Then $g \in \mathbb{N}$ and $\forall b \in \mathbb{N}(g < b \lor g = b \lor b < g)$. $s(g) \in \mathbb{N}$ exists. Let arbitrary $b \in \mathbb{N}$. We have 3 cases. First suppose $g < b$. Then there exists $x \in \mathbb{N}$ such that $g + x = b$. Suppose $x = 1$; then $s(g) = g + 1 = b$. Next suppose $x \neq 1$; then by lemma 1.3 there exists $y \in \mathbb{N}$ such that $x = s(y)$. So $b = g + x = g + s(y) = g + (y + 1)$. By theorem 1.7 parts 2 and 4 we have $(g+1)+y = b$ and $s(g) < b$. For the second case suppose $g = b$. Then by (1) we have $b = g < g + 1 = s(g)$. Finally suppose $b < g$. Then since $g < g + 1$ by (3) we have $b < g + 1 = s(g)$. We then have $G = \mathbb{N}$.

Showing the desired result is straightforward. For all arbitrary $a \in \mathbb{N}$, $a \in G$. The result immediately follows. □

Proof. (6) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N}(ac < bc \leftrightarrow a < b)$$

Let arbitrary $a, b, c \in \mathbb{N}$. First suppose $a < b$. Then there exists $x \in \mathbb{N}$ such that $a + x = b$. Then by theorem 1.7 part 8 we have $bc = (a + x)c = (ac) + (xc)$; so $ac < bc$.

Next suppose $ac < bc$. Suppose for contradiction that $a \not< b$. Then by (5) we have $a = b \lor b < a$. First suppose $a = b$. Then we immediately have $ac < ac$, which contradicts (1). Next suppose $b < a$. There exists $x, y \in \mathbb{N}$ such that $(ac) + x = bc$ and $b + y = a$. By theorem 1.7 parts 8 and 2 we have $bc = ((b + y)c) + x = ((bc) + (yc)) + x = (bc) + ((yc) + x)$, contradicting part 6 of that same theorem. □

(next page)

Proof. (7) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(a \leq b \vee b \leq a)$$

Let arbitrary $a, b \in \mathbb{N}$. First suppose $a \not< b$. Then by (5) $a = b \vee b < a$, meaning $b \leq a$. Next suppose $a < b$. Then immediately it follows that $a \leq b$. □

Proof. (8) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}((a \leq b \wedge b \leq a) \rightarrow a = b)$$

Let arbitrary $a, b \in \mathbb{N}$. Suppose that $a \leq b \wedge b \leq a$. We have four cases. Three of the cases: $a < b \wedge b < a$, $a = b \wedge b < a$, and $a < b \wedge b = a$ all contradict (5). The only possibility is the last case $a = b \wedge b = a$. □

Proof. (9) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(\neg(b < a \wedge a < b + 1))$$

Let arbitrary $a, b \in \mathbb{N}$. Suppose for contradiction that $b < a \wedge a < b + 1$. Then there exists $x, y \in \mathbb{N}$ such that $b+x = a$ and $a+y = b+1$. Then $(b+x)+y = b+1$. By theorem 1.7 parts 2 and 4 we have $(x + y) + b = 1 + b$. By part 1 of that theorem we then have $x + y = 1$, contradicting part 5 of that theorem. □

Proof. (10) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(a \leq b \leftrightarrow a < b + 1)$$

Let arbitrary $a, b \in \mathbb{N}$. First suppose $a \leq b$. Suppose for contradiction that $b + 1 \leq a$. Then by (3) we would have $b + 1 \leq b$. The case where $b + 1 = b$ is a contradiction to theorem 1.7 part 6. The case where $b + 1 < b$ implies the existence of $x \in \mathbb{N}$ such that $(b + 1) + x = b$; by theorem 1.7 part 2 we have $b + (1 + x) = b$, again contradicting part 6 of that theorem. We can then conclude that $\neg(b + 1 \leq a)$, meaning $\neg(b + 1 = a \vee b + 1 < a)$. By (5) we must then have $a < b + 1$.

Now for the other direction. Suppose $a < b + 1$. Supposing that $b < a$ immediately contradicts (9). Therefore by (5) we must have $a \leq b$. □

Proof. (11) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(a < b \leftrightarrow a + 1 \leq b)$$

Let arbitrary $a, b \in \mathbb{N}$. Suppose $a < b$. Supposing $b < a + 1$ immediately contradicts (9), so we must have, by (5), $a + 1 \leq b$.

Next suppose $a + 1 \leq b$. Suppose for contradiction that $b \leq a$, which would imply by (3) that $a + 1 \leq a$. The case where $a + 1 = a$ immediately contradicts theorem 1.7 part 6. The case where $a + 1 < a$ implies the existence of $x \in \mathbb{N}$ such that $(a+1)+x = a$; theorem 1.7 part 2 then implies $a+(1+x) = a$, again contradicting part 6 of that same theorem. We conclude that $\neg(b = a \vee b < a)$, so by (5) we have $a < b$. □

1.1.7 Well-ordering

Theorem 1.11. *(Well-Ordering Principle) Let $G \subseteq \mathbb{N}$ be a non-empty set. Then there is some $m \in G$ such that $m \leq g$ for all $g \in G$.*

Proof. We want to show

$$\forall G(G \subseteq \mathbb{N} \wedge G \neq \emptyset \to \exists m \in G(\forall g \in G(m \leq g)))$$

Let G be arbitrary; assume that $G \subseteq \mathbb{N}$ and $G \neq \emptyset$. Assume for contradiction that the implication is false. That is

$$\neg \exists m \in G(\forall g \in G(m \leq g))$$

We define H as the set such that

$$\forall a(a \in H \leftrightarrow a \in \mathbb{N} \wedge \forall n \in \mathbb{N}(n \leq a \to n \notin G))$$

(where existence and uniqueness can be easily shown). We can show that $H \cap G = \emptyset$. For contradiction suppose there exists x such that $x \in H$ and $x \in G$. By definition then $x \in \mathbb{N}$ and $\forall n \in \mathbb{N}(n \leq x \to n \notin G)$. Then since $x \leq x$ we must have $x \notin G$, a contradiction. We will show that $H = \mathbb{N}$. Showing $H \subseteq \mathbb{N}$ is straightforward. We need to show $1 \in H$ and $\forall h \in H(s(h) \in H)$.

We start with the former. Suppose for contradiction that $1 \notin H$. We know $1 \in \mathbb{N}$, so we must have $\neg \forall n \in \mathbb{N}(n \leq 1 \to n \notin G)$. This is logically equivalent to

$$\exists n \in \mathbb{N}(n \leq 1 \wedge n \in G)$$

That is, there exists $n \in \mathbb{N}$ such that $n \leq 1$ and $n \in G$. $n < 1$ is a contradiction as it would imply the existence of $x \in \mathbb{N}$ such that $n + x = 1$, contradicting theorem 1.7 part 5. We therefore must have $1 = n \in G$. Let arbitrary $g \in G \subseteq \mathbb{N}$. By theorem 1.10 part 2 we must have $n = 1 \leq g$. This contradicts our assumption that such an element does not exist, so $1 \in H$.

Now the latter. Let arbitary $h \in H$. Then $h \in \mathbb{N}$ and $\forall n \in \mathbb{N}(n \leq h \to n \notin G)$. We know $s(h) \in \mathbb{N}$ exists. Suppose for contradiction that $s(h) \notin H$. Then as before there then exists $n \in \mathbb{N}$ such that $n \leq s(h)$ and $n \in G$. By the contrapositive of one of the consequences of $h \in H$, since $n \in G$ we then have $\neg(n \leq h)$, which by theorem 1.10 part 5 implies $h < n$. Since $n < s(h) = h + 1$ would then lead to a contradiction of theorem 1.10 part 9, we must have $n = s(h)$. Let arbitrary $g \in G$. Suppose for contradiction that $g < n = s(h) = h+1$. Then by theorem 1.10 part 10 we have $g \leq h$. Since $h \in H$ we then have $g \notin G$, a contradiction. So we must have $\neg(g < n)$, which by theorem 1.10 part 5 implies $n \leq g$. We know that $n \in G$, so this contradicts our earlier assumption that such an element must not exist. So $s(h) \in H$ and $H = \mathbb{N}$.

We know $H \cap G = \emptyset$. So $\mathbb{N} \cap G = \emptyset$. Suppose for contradiction that G is nonempty, meaning that there exists some $x \in G$. Then since $G \subseteq \mathbb{N}$ we would have $x \in \mathbb{N}$ and therefore $x \in \mathbb{N} \cap G = \emptyset$, a contradiction. So we must have $G = \emptyset$, contradicting a key assumption. $\square$

1.2 Construction of the Integers

1.2.1 Construction

Definition 1.12. *The relation* $\sim$ *on* $\mathbb{N} \times \mathbb{N}$ *is defined by* $\langle a, b\rangle \sim \langle c, d\rangle$ *if and only if* $a + d = b + c$ *for all* $\langle a, b\rangle, \langle c, d\rangle \in \mathbb{N} \times \mathbb{N}$. *That is, the set such that*

$$\forall x(x \in \sim \leftrightarrow \exists \langle a, b\rangle \in \mathbb{N} \times \mathbb{N} \exists \langle c, d\rangle \in \mathbb{N} \times \mathbb{N}(x = \langle\langle a, b\rangle, \langle c, d\rangle\rangle \wedge a + d = b + c))$$

(Existence follows from using $(\mathbb{N} \times \mathbb{N}) \times (\mathbb{N} \times \mathbb{N})$ as a bounding set. Uniqueness from extensionality)

Lemma 1.13. *The relation* $\sim$ *is an equivalence relation on* $\mathbb{N} \times \mathbb{N}$.

Proof. Showing that $\sim \subseteq (\mathbb{N} \times \mathbb{N}) \times (\mathbb{N} \times \mathbb{N})$ is straightforward. We need to show reflexivity on $\mathbb{N} \times \mathbb{N}$, symmetry, and transitivity. We start with reflexivity; let arbitrary $\langle a, b\rangle \in \mathbb{N} \times \mathbb{N}$. We immediately have $a + b = b + a$ and $\langle a, b\rangle \sim \langle a, b\rangle$. For symmetry, let $\langle a, b\rangle, \langle c, d\rangle$ be arbitrary and suppose $\langle a, b\rangle \sim \langle c, d\rangle$. Then by definition both pairs are members of $\mathbb{N} \times \mathbb{N}$ and $a + d = b + c$. We then have $c + b = b + c = a + d = d + a$, which then means $\langle c, d\rangle \sim \langle a, b\rangle$.

Finally transitivity. Let $\langle a, b\rangle, \langle c, d\rangle, \langle e, f\rangle$ be arbitrary, suppose $\langle a, b\rangle \sim \langle c, d\rangle$ and $\langle c, d\rangle \sim \langle e, f\rangle$. This means that all three pairs are members of $\mathbb{N} \times \mathbb{N}$, $a + d = c + b$, and $c + f = e + d$. We then have

$$\begin{aligned}(a + f) + c &= a + (c + f) = a + (e + d) = (a + d) + e \\ &= (c + b) + e = (b + e) + c\end{aligned}$$

By cancellation we then have $a + f = b + e$. So $\langle a, b\rangle \sim \langle e, f\rangle$. $\square$

Definition 1.14. *The set of* ***integers****, denoted* $\mathbb{Z}$*, is the quotient set* $(\mathbb{N} \times \mathbb{N})/\sim$. *That is,*

$$\forall x(x \in \mathbb{Z} \leftrightarrow \exists \langle a, b\rangle \in \mathbb{N} \times \mathbb{N}(x = [\langle a, b\rangle]))$$

where

$$\forall \langle a, b\rangle \in \mathbb{N} \times \mathbb{N} \forall x(x \in [\langle a, b\rangle] \leftrightarrow \langle a, b\rangle \sim x)$$

Each integer is an equivalence class. Existence of each equivalence class can be shown using $\mathrm{ran}(\sim)$ as a bounding set. The same can be done for $\mathbb{Z}$ using $\mathscr{P}(\mathrm{ran}(\sim))$. Uniqueness for both follows from extensionality.

Definition 1.15. *We define* $\hat{0}, \hat{1}$ *as* $\hat{0} = [\langle 1, 1\rangle]$ *and* $\hat{1} = [\langle 1 + 1, 1\rangle]$.

1.2.2 Operations on the integers

Definition 1.16. *The binary operation* $+$ *on* $\mathbb{Z}$ *is defined as* $+ : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ *where*

$$\forall x(x \in + \leftrightarrow \exists \langle a,b\rangle \in \mathbb{N}\times\mathbb{N} \exists \langle c,d\rangle \in \mathbb{N}\times\mathbb{N}(x = \langle\langle[\langle a,b\rangle],[\langle c,d\rangle]\rangle,[\langle a+c,b+d\rangle]\rangle))$$

put simply, for all $[\langle a,b\rangle],[\langle c,d\rangle] \in \mathbb{Z}$*,*

$$[\langle a,b\rangle] + [\langle c,d\rangle] = [\langle a+c, b+d\rangle]$$

(A bounding set is $(\mathbb{Z}\times\mathbb{Z})\times\mathbb{Z}$. Uniqueness follows from extensionality.)

Proof. We start by showing the domain bound $\mathrm{dom}(+) = \mathbb{Z}\times\mathbb{Z}$. First suppose $x \in \mathrm{dom}(+)$. Then there exists y such that $\langle x,y\rangle \in +$ and $\langle a,b\rangle,\langle c,d\rangle \in \mathbb{N}\times\mathbb{N}$ such that $x = \langle[\langle a,b\rangle],[\langle c,d\rangle]\rangle \in \mathbb{Z}\times\mathbb{Z}$. Next suppose $x \in \mathbb{Z}\times\mathbb{Z}$. Then there exists $x_0, x_1 \in \mathbb{Z}$ such that $x = \langle x_0, x_1\rangle$. There exists $\langle a,b\rangle,\langle c,d\rangle \in \mathbb{N}\times\mathbb{N}$ such that $x_0 = [\langle a,b\rangle]$ and $x_1 = [\langle c,d\rangle]$. We know $\langle a+c, b+d\rangle \in \mathbb{N}\times\mathbb{N}$ and $\langle\langle[\langle a,b\rangle],[\langle c,d\rangle]\rangle,[\langle a+c,b+d\rangle]\rangle \in +$, so $x = \langle x_0,x_1\rangle = \langle[\langle a,b\rangle],[\langle c,d\rangle]\rangle \in \mathrm{dom}(+)$.

Now the range bound $\mathrm{ran}(+) \subseteq \mathbb{Z}$. Let arbitrary $y \in \mathrm{ran}(+)$. Then there exists x such that $\langle x,y\rangle \in +$ and $a,b,c,d \in \mathbb{N}$ such that $y = [\langle a+c,b+d\rangle] \in \mathbb{Z}$.

Now we prove well definedness. That is, that addition between equivalence classes is independent of the particular elements used to represent each equivalence class:

$$\begin{aligned}
&\forall \langle a,b\rangle \in \mathbb{N}\times\mathbb{N} \forall \langle c,d\rangle \in \mathbb{N}\times\mathbb{N} \forall \langle a',b'\rangle \in \mathbb{N}\times\mathbb{N} \forall \langle c',d'\rangle \in \mathbb{N}\times\mathbb{N}(\\
&\qquad [\langle a,b\rangle] = [\langle a',b'\rangle] \land [\langle c,d\rangle] = [\langle c',d'\rangle] \\
&\qquad\qquad \to [\langle a+c,b+d\rangle] = [\langle a'+c',b'+d'\rangle])
\end{aligned}$$

Let arbitary $\langle a,b\rangle,\langle c,d\rangle,\langle a',b'\rangle,\langle c',d'\rangle \in \mathbb{N}\times\mathbb{N}$ and suppose that $[\langle a,b\rangle] = [\langle a',b'\rangle]$ and $[\langle c,d\rangle] = [\langle c',d'\rangle]$. Then $\langle a,b\rangle \sim \langle a',b'\rangle$ and $a+b' = b+a'$. Similarly $c+d' = d+c'$. We then have, by associativity and commutativity of natural arithmetic:

$$\begin{aligned}
(a+c)+(b'+d') &= (a+b')+(c+d') \\
&= (b+a')+(d+c') = (b+d)+(a'+c')
\end{aligned}$$

and we can conclude that $\langle a+c,b+d\rangle \sim \langle a'+c',b'+d'\rangle$ and $[\langle a+c,b+d\rangle] = [\langle a'+c',b'+d'\rangle]$.

Finally we show functionality. Let arbitrary $x \in \mathrm{dom}(+)$. Then by definition there exists y such that $\langle x,y\rangle \in +$. There then exists $\langle a,b\rangle,\langle c,d\rangle \in \mathbb{N}\times\mathbb{N}$ such that $x = \langle[\langle a,b\rangle],[\langle c,d\rangle]\rangle$ and $y = [\langle a+c,b+d\rangle]$. Let y' be arbitrary such that $\langle x,y'\rangle \in +$. Then similarly there exists $\langle a',b'\rangle,\langle c',d'\rangle \in \mathbb{N}\times\mathbb{N}$ such that $x = \langle[\langle a',b'\rangle],[\langle c',d'\rangle]\rangle$ and $y' = [\langle a'+c',b'+d'\rangle]$. We know $\langle[\langle a,b\rangle],[\langle c,d\rangle]\rangle = \langle[\langle a',b'\rangle],[\langle c',d'\rangle]\rangle$. So since addition is well defined we have $y = [\langle a+c,b+d\rangle] = [\langle a'+c',b'+d'\rangle] = y'$. ∎

(next page)

Definition 1.17. *The binary operation* $\cdot$ *on* $\mathbb{Z}$ *is defined as* $\cdot : \mathbb{Z}\times\mathbb{Z} \to \mathbb{Z}$ *where*

$$\forall x(x \in \cdot \leftrightarrow \exists\langle a,b\rangle \in \mathbb{N}\times\mathbb{N}\exists\langle c,d\rangle \in \mathbb{N}\times\mathbb{N}(\\ x = \langle\langle[\langle a,b\rangle],[\langle c,d\rangle]\rangle,[\langle ac+bd, ad+bc\rangle]\rangle))$$

(A bounding set is $(\mathbb{Z}\times\mathbb{Z})\times\mathbb{Z}$. Uniqueness follows from extensionality.)

Proof. We start with the domain bound $\mathrm{dom}(\cdot) = \mathbb{Z}\times\mathbb{Z}$. First suppose $x \in \mathrm{dom}(\cdot)$. Then there exists y such that $\langle x,y\rangle \in \cdot$, and $\langle a,b\rangle, \langle c,d\rangle \in \mathbb{N}\times\mathbb{N}$ such that $x = \langle[\langle a,b\rangle],[\langle c,d\rangle]\rangle \in \mathbb{Z}\times\mathbb{Z}$. Next suppose $x \in \mathbb{Z}\times\mathbb{Z}$. Then there exists $x_0, x_1 \in \mathbb{Z}$ such that $x = \langle x_0, x_1\rangle$. There exists $\langle a,b\rangle, \langle c,d\rangle \in \mathbb{N}\times\mathbb{N}$ such that $x_0 = [\langle a,b\rangle]$ and $x_1 = [\langle c,d\rangle]$. We have $\langle\langle[\langle a,b\rangle],[\langle c,d\rangle]\rangle,[\langle ac+bd, ad+bc\rangle]\rangle \in \cdot$. So $x = \langle x_0,x_1\rangle = \langle[\langle a,b\rangle],[\langle c,d\rangle]\rangle \in \mathrm{dom}(\cdot)$.

Next the range bound $\mathrm{ran}(\cdot) \subseteq \mathbb{Z}$. Let arbitary $y \in \mathrm{ran}(\cdot)$. Then there exists x such that $\langle x,y\rangle \in \cdot$ and $a,b,c,d \in \mathbb{N}$ such that $y = [\langle ac+bd, ad+bc\rangle] \in \mathbb{Z}$.

Next we show well definedness. That is

$$\forall\langle a,b\rangle \in \mathbb{N}\times\mathbb{N}\forall\langle c,d\rangle \in \mathbb{N}\times\mathbb{N}\forall\langle a',b'\rangle \in \mathbb{N}\times\mathbb{N}\forall\langle c',d'\rangle \in \mathbb{N}\times\mathbb{N}(\\ ([\langle a,b\rangle] = [\langle a',b'\rangle] \wedge [\langle c,d\rangle] = [\langle c',d'\rangle]) \\ \to [\langle ac+bd, ad+bc\rangle] = [\langle a'c'+b'd', a'd'+b'c'\rangle])$$

Let arbitrary $\langle a,b\rangle, \langle c,d\rangle, \langle a',b'\rangle, \langle c',d'\rangle \in \mathbb{N}\times\mathbb{N}$ and suppose $[\langle a,b\rangle] = [\langle a',b'\rangle]$ and $[\langle c,d\rangle] = [\langle c',d'\rangle]$. Then we have $\langle a,b\rangle \sim \langle a',b'\rangle$ and $a+b' = b+a'$. Similarly we have $c+d' = d+c'$. We want to show

$$(ac+bd) + (a'd'+b'c') = (ad+bc) + (a'c'+b'd')$$

See that

$$\begin{aligned} &((a+b')d + (b+a')c) + ((c+d')b' + (d+c')a') \\ &= ((ad+b'd) + (bc+a'c)) + ((cb'+d'b') + (da'+c'a')) \\ &= ((ad+bc) + (b'd+a'c)) + ((a'c'+b'd') + (da'+cb')) \\ &= ((ad+bc) + (a'c'+b'd')) + ((b'd+a'c) + (da'+cb')) \end{aligned}$$

Also see that

$$\begin{aligned} &((a+b')d + (b+a')c) + ((c+d')b' + (d+c')a') \\ &= ((b+a')d + (a+b')c) + ((d+c')b' + (c+d')a') \\ &= ((bd+a'd) + (ac+b'c)) + ((db'+c'b') + (ca'+d'a')) \\ &= ((ac+bd) + (da'+cb')) + ((a'd'+c'b') + (a'c+b'd)) \\ &= ((ac+bd) + (a'd'+c'b')) + ((b'd+a'c) + (da'+cb')) \end{aligned}$$

By cancellation we then have our desired result, which means

$$\langle ac+bd, ad+bc\rangle \sim \langle a'c'+b'd', a'd'+b'c'\rangle$$

and $[\langle ac+bd, ad+bc\rangle] = [\langle a'c'+b'd', a'd'+b'c'\rangle]$.
(next page)

Finally functionality. Let arbitrary $x \in \text{dom}(\cdot)$. Then there exists y such that $\langle x, y\rangle \in \cdot$ and $\langle a, b\rangle, \langle c, d\rangle \in \mathbb{N} \times \mathbb{N}$ such that $x = \langle [\langle a, b\rangle], [\langle c, d\rangle]\rangle$ and $y = [\langle ac + bd, ad + bc\rangle]$. Let y' be arbitrary such that $\langle x, y'\rangle \in \cdot$. Then similarly there exists $\langle a', b'\rangle, \langle c', d'\rangle \in \mathbb{N} \times \mathbb{N}$ such that $x = \langle [\langle a', b'\rangle], [\langle c', d'\rangle]\rangle$ and $y' = [\langle a'c' + b'd', a'd' + b'c'\rangle]$. We have $\langle [\langle a, b\rangle], [\langle c, d\rangle]\rangle = \langle [\langle a', b'\rangle], [\langle c', d'\rangle]\rangle$, by well definedness we then have $y = [\langle ac + bd, ad + bc\rangle] = [\langle a'c' + b'd', a'd' + b'c'\rangle] = y'$. □

Definition 1.18. *The unary operation $-$ on $\mathbb{Z}$ is defined as $- : \mathbb{Z} \to \mathbb{Z}$ where*

$$\forall x(x \in - \leftrightarrow \exists \langle a, b\rangle \in \mathbb{N} \times \mathbb{N}(x = \langle [\langle a, b\rangle], [\langle b, a\rangle]\rangle))$$

(A bounding set is $\mathbb{Z} \times \mathbb{Z}$. Uniqueness follows from extensionality.)

Proof. We start with the domain bound $\text{dom}(-) = \mathbb{Z}$. First let arbitrary $x \in \text{dom}(-)$. Then there exists y such that $\langle x, y\rangle \in -$, and $\langle a, b\rangle \in \mathbb{N} \times \mathbb{N}$ such that $x = [\langle a, b\rangle] \in \mathbb{Z}$. Next suppose $x \in \mathbb{Z}$. Then there exists $\langle a, b\rangle \in \mathbb{N} \times \mathbb{N}$ such that $x = [\langle a, b\rangle]$. We have $\langle [\langle a, b\rangle], [\langle b, a\rangle]\rangle \in -$. So $x = [\langle a, b\rangle] \in \text{dom}(-)$.

Next the range bound $\text{ran}(-) \subseteq \mathbb{Z}$. Suppose $y \in \text{ran}(-)$. Then there exists x such that $\langle x, y\rangle \in -$ and $a, b \in \mathbb{N}$ such that $y = [\langle b, a\rangle] \in \mathbb{Z}$.

Next we show well-defineness. That is

$$\forall \langle a, b\rangle \in \mathbb{N} \times \mathbb{N} \forall \langle a', b'\rangle \in \mathbb{N} \times \mathbb{N}([\langle a, b\rangle] = [\langle a', b'\rangle] \to [\langle b, a\rangle] = [\langle b', a'\rangle])$$

Letting arbitrary $\langle a, b\rangle, \langle a', b'\rangle \in \mathbb{N} \times \mathbb{N}$ and supposing that $[\langle a, b\rangle] = [\langle a', b'\rangle]$, we have $a + b' = b + a'$. See that this also means $\langle b, a\rangle \sim \langle b', a'\rangle$ and therefore $[\langle b, a\rangle] = [\langle b', a'\rangle]$.

Finally functionality. Let arbitary $x \in \text{dom}(-)$. Then there exists y such that $\langle x, y\rangle \in -$, and $\langle a, b\rangle \in \mathbb{N} \times \mathbb{N}$ such that $x = [\langle a, b\rangle]$ and $y = [\langle b, a\rangle]$. Let y' be arbitrary such that $\langle x, y'\rangle \in -$. Then there exists $\langle a', b'\rangle \in \mathbb{N} \times \mathbb{N}$ such that $x = [\langle a', b'\rangle]$ and $y' = [\langle b', a'\rangle]$. We have $[\langle a, b\rangle] = [\langle a', b'\rangle]$; by well-definedness we have $y = [\langle b, a\rangle] = [\langle b', a'\rangle] = y'$. □

1.2.3 Ordering

Definition 1.19. *The binary relation $<$ on $\mathbb{Z}$ is defined such that*

$$\forall x(x \in < \leftrightarrow \exists \langle a,b\rangle \in \mathbb{N}\times\mathbb{N}\exists\langle c,d\rangle \in \mathbb{N}\times\mathbb{N}(x = \langle[\langle a,b\rangle],[\langle c,d\rangle]\rangle \wedge a+d < b+c))$$

($\mathbb{Z}\times\mathbb{Z}$ is a bounding set. Uniqueness follows from extensionality.)

Lemma 1.20.

$$\begin{gathered}\forall\langle a,b\rangle \in \mathbb{N}\times\mathbb{N}\forall\langle c,d\rangle \in \mathbb{N}\times\mathbb{N}\forall\langle a',b'\rangle \in \mathbb{N}\times\mathbb{N}\forall\langle c',d'\rangle \in \mathbb{N}\times\mathbb{N}(\\ ([\langle a,b\rangle] = [\langle a',b'\rangle] \wedge [\langle c,d\rangle] = [\langle c',d'\rangle]) \\ \to a+d < b+c \leftrightarrow a'+d' < b'+c')\end{gathered}$$

Proof. Let arbitrary $\langle a,b\rangle, \langle c,d\rangle, \langle a',b'\rangle, \langle c',d'\rangle \in \mathbb{N}\times\mathbb{N}$. Suppose that $[\langle a,b\rangle] = [\langle a',b'\rangle]$ and $[\langle c,d\rangle] = [\langle c',d'\rangle]$, which then means $a+b' = b+a'$ and $c+d' = d+c'$. First suppose $a+d < b+c$. Then there exists x such that $(a+d)+x = b+c$, and we have

$$(((a+d)+x)+(b+a'))+(c+d') = ((b+c)+(a+b'))+(d+c')$$

Rearranging the left side:

$$\begin{aligned}&(((a+d)+x)+(b+a'))+(c+d')\\ &= ((a+d)+x)+((b+c)+(a'+d'))\\ &= (a+d)+(x+((b+c)+(a'+d')))\\ &= (a+d)+((b+c)+(x+(a'+d')))\\ &= ((a+d)+(b+c))+((a'+d')+x)\end{aligned}$$

and the right side:

$$\begin{aligned}&((b+c)+(a+b'))+(d+c')\\ &= (b+c)+((a+b')+(d+c'))\\ &= (b+c)+((a+d)+(b'+c'))\\ &= ((a+d)+(b+c))+(b'+c')\end{aligned}$$

By cancellation we have $(a'+d')+x = b'+c'$ and $a'+d' < b'+c'$.
(next page)

Next suppose $a'+d' < b'+c'$. Then there exists x such that $a'+d'+x = b'+c'$. We have

$$(((a' + d') + x) + (a + b')) + (d + c') = ((b' + c') + (b + a')) + (c + d')$$

Rearranging the left side:

$$\begin{aligned}
&(((a' + d') + x) + (a + b')) + (d + c') \\
&= ((a' + d') + x) + ((a + b') + (d + c')) \\
&= ((a' + d') + x) + ((b' + c') + (a + d)) \\
&= (a' + d') + (x + ((b' + c') + (a + d))) \\
&= ((a' + d') + (b' + c')) + ((a + d) + x)
\end{aligned}$$

and the right hand side:

$$\begin{aligned}
&((b' + c') + (b + a')) + (c + d') \\
&= (b' + c') + ((b + a') + (c + d')) \\
&= (b' + c') + ((a' + d') + (b + c)) \\
&= ((a' + d') + (b' + c')) + (b + c)
\end{aligned}$$

So by cancellation $(a + d) + x = b + c$ and $a + d < b + c$. □

Lemma 1.21.

$$\forall\langle a, b\rangle \in \mathbb{N} \times \mathbb{N}\forall\langle c, d\rangle \in \mathbb{N} \times \mathbb{N}([\langle a, b\rangle] < [\langle c, d\rangle] \leftrightarrow a + d < b + c)$$

Proof. Let arbitrary $\langle a, b\rangle, \langle c, d\rangle \in \mathbb{N} \times \mathbb{N}$. First suppose $[\langle a, b\rangle] < [\langle c, d\rangle]$. Then by definition there exists $\langle a_0, b_0\rangle, \langle c_0, d_0\rangle \in \mathbb{N} \times \mathbb{N}$ such that $[\langle a, b\rangle] = [\langle a_0, b_0\rangle]$, $[\langle c, d\rangle] = [\langle c_0, d_0\rangle]$ and $a_0 + d_0 < b_0 + c_0$. It then immediately follows by lemma 1.20 that $a + d < b + c$.

The reverse direction is straightforward. Suppose $a + d < b + c$ and it immediately follows by definition that $[\langle a, b\rangle] < [\langle c, d\rangle]$. □

Proof. (range/domain bound and well-definedness) We start by showing $<$ is a relation on $\mathbb{Z}$, $<\subseteq \mathbb{Z} \times \mathbb{Z}$. Let arbitary $x \in<$. Then by definition there exists $\langle a, b\rangle, \langle c, d\rangle \in \mathbb{N} \times \mathbb{N}$ such that $x = \langle[\langle a, b\rangle], [\langle c, d\rangle]\rangle \in \mathbb{Z} \times \mathbb{Z}$.

Next well-definedness. That is,

$$\begin{aligned}
&\forall\langle a, b\rangle \in \mathbb{N} \times \mathbb{N}\forall\langle c, d\rangle \in \mathbb{N} \times \mathbb{N}\forall\langle a', b'\rangle \in \mathbb{N} \times \mathbb{N}\forall\langle c', d'\rangle \in \mathbb{N} \times \mathbb{N}(\\
&\qquad([\langle a, b\rangle] = [\langle a', b'\rangle] \wedge [\langle c, d\rangle] = [\langle c', d'\rangle]) \\
&\qquad\to ([\langle a, b\rangle] < [\langle c, d\rangle] \leftrightarrow [\langle a', b'\rangle] < [\langle c', d'\rangle]))
\end{aligned}$$

Letting arbitary $\langle a, b\rangle, \langle c, d\rangle, \langle a', b'\rangle, \langle c', d'\rangle \in \mathbb{N} \times \mathbb{N}$. Suppose that $[\langle a, b\rangle] = [\langle a', b'\rangle]$ and $[\langle c, d\rangle] = [\langle c', d'\rangle]$, which then means $a+b' = b+a'$ and $c+d' = d+c'$. Suppose first that $[\langle a, b\rangle] < [\langle c, d\rangle]$, which by lemma 1.21 means $a+d < b+c$. It follows by lemma 1.20 that $a' + d' < b' + c'$ and by lemma 1.21 that $[\langle a', b'\rangle] < [\langle c', d'\rangle]$. An identical strategy proves the reverse direction. □

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Definition 1.22. *The binary relation* $\leq$ *on* $\mathbb{Z}$ *is defined such that*

$$\forall x(x \in \leq \leftrightarrow \exists\langle a,b\rangle \in \mathbb{N}\times\mathbb{N}\exists\langle c,d\rangle \in \mathbb{N}\times\mathbb{N}(x = \langle[\langle a,b\rangle],[\langle c,d\rangle]\rangle \\ \wedge ([\langle a,b\rangle] < [\langle c,d\rangle] \vee [\langle a,b\rangle] = [\langle c,d\rangle])))$$

($\mathbb{Z}\times\mathbb{Z}$ is a bounding set. Uniqueness follows from extensionality.)

Lemma 1.23.

$$\forall\langle a,b\rangle \in \mathbb{N}\times\mathbb{N}\forall\langle c,d\rangle \in \mathbb{N}\times\mathbb{N}([\langle a,b\rangle] \leq [\langle c,d\rangle] \\ \leftrightarrow ([\langle a,b\rangle] < [\langle c,d\rangle] \vee [\langle a,b\rangle] = [\langle c,d\rangle]))$$

Proof. Let arbitrary $\langle a,b\rangle,\langle c,d\rangle \in \mathbb{N}\times\mathbb{N}$. First suppose $[\langle a,b\rangle] \leq [\langle c,d\rangle]$. Then by definition there exists $\langle a_0,b_0\rangle,\langle c_0,d_0\rangle \in \mathbb{N}\times\mathbb{N}$ such that $[\langle a,b\rangle] = [\langle a_0,b_0\rangle]$, $[\langle c,d\rangle] = [\langle c_0,d_0\rangle]$, and $[\langle a_0,b_0\rangle] < [\langle c_0,d_0\rangle]$ or $[\langle a_0,b_0\rangle] = [\langle c_0,d_0\rangle]$. In the first case suppose $[\langle a_0,b_0\rangle] < [\langle c_0,d_0\rangle]$. It then follows from the well-definedness of $<$ that $[\langle a,b\rangle] < [\langle c,d\rangle]$. In the other case suppose $[\langle a_0,b_0\rangle] = [\langle c_0,d_0\rangle]$. It immediately follows that $[\langle a,b\rangle] = [\langle a_0,b_0\rangle] = [\langle c_0,d_0\rangle] = [\langle c,d\rangle]$.

Now suppose $[\langle a,b\rangle] < [\langle c,d\rangle] \vee [\langle a,b\rangle] = [\langle c,d\rangle]$. It immediately follows by definition that $[\langle a,b\rangle] \leq [\langle c,d\rangle]$. □

Proof. (domain/range bound and well-definedness) First we show $\leq \subseteq \mathbb{Z}\times\mathbb{Z}$. Let arbitrary $x \in \leq$. Then there exists $\langle a,b\rangle,\langle c,d\rangle \in \mathbb{N}\times\mathbb{N}$ such that $x = \langle[\langle a,b\rangle],[\langle c,d\rangle]\rangle \in \mathbb{Z}\times\mathbb{Z}$.

Next well-definedness. That is

$$\forall\langle a,b\rangle \in \mathbb{N}\times\mathbb{N}\forall\langle c,d\rangle \in \mathbb{N}\times\mathbb{N}\forall\langle a',b'\rangle \in \mathbb{N}\times\mathbb{N}\forall\langle c',d'\rangle \in \mathbb{N}\times\mathbb{N}(\\ ([\langle a,b\rangle] = [\langle a',b'\rangle] \wedge [\langle c,d\rangle] = [\langle c',d'\rangle]) \\ \to ([\langle a,b\rangle] \leq [\langle c,d\rangle] \leftrightarrow [\langle a',b'\rangle] \leq [\langle c',d'\rangle]))$$

Letting arbitary $\langle a,b\rangle,\langle c,d\rangle,\langle a',b'\rangle,\langle c',d'\rangle \in \mathbb{N}\times\mathbb{N}$. Suppose that $[\langle a,b\rangle] = [\langle a',b'\rangle]$ and $[\langle c,d\rangle] = [\langle c',d'\rangle]$. First suppose $[\langle a,b\rangle] \leq [\langle c,d\rangle]$. It follows by lemma 1.23 that $[\langle a,b\rangle] < [\langle c,d\rangle] \vee [\langle a,b\rangle] = [\langle c,d\rangle]$. In the first case where $[\langle a,b\rangle] < [\langle c,d\rangle]$ it follows from the well-definedness of $<$ that $[\langle a',b'\rangle] < [\langle c',d'\rangle]$ and again by lemma 1.23 that $[\langle a',b'\rangle] \leq [\langle c',d'\rangle]$. For the other case where $[\langle a,b\rangle] = [\langle c,d\rangle]$ we immediately have $[\langle a',b'\rangle] = [\langle a,b\rangle] = [\langle c,d\rangle] = [\langle c',d'\rangle]$ and by lemma 1.23 $[\langle a',b'\rangle] \leq [\langle c',d'\rangle]$. The reverse implication follows by the same argument. □

1.2.4 Properties

Theorem 1.24. *Let $x, y, z \in \mathbb{Z}$.*

1. $(x+y)+z = x+(y+z)$ *(Associative Law for Addition)*
2. $x+y = y+x$ *(Commutative Law for Addition)*
3. $x+\hat{0} = x$ *(Identity Law for Addition)*
4. $x+(-x) = \hat{0}$ *(Inverses Law for Addition)*
5. $(xy)z = x(yz)$ *(Associative Law for Multiplication)*
6. $xy = yx$ *(Commutative Law for Multiplication)*
7. $x \cdot \hat{1} = x$ *(Identity Law for Multiplication)*
8. $x(y+z) = xy+xz$ *(Distributive Law)*
9. *If* $xy = \hat{0}$, *then* $x = \hat{0}$ *or* $y = \hat{0}$ *(No Zero Divisors Law)*
10. *Precisely one of* $x < y$ *or* $x = y$ *or* $x > y$ *holds* *(Trichotomy Law)*
11. *If* $x < y$ *and* $y < z$, *then* $x < z$ *(Transitive Law)*
12. *If* $x < y$ *then* $x+z < y+z$ *(Addition Law for Order)*
13. *If* $x < y$ *and* $z > \hat{0}$, *the* $xz < yz$ *(Multiplication Law for Order)*
14. $\hat{0} \neq \hat{1}$ *(Non-Triviality)*

Proof. (1) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} \forall z \in \mathbb{Z}((x+y)+z = x+(y+z))$$

Let arbitrary $x, y, z \in \mathbb{Z}$. Then there exists $a, b, c, d, e, f \in \mathbb{N}$ such that $x = [\langle a, b\rangle]$, $y = [\langle c, d\rangle]$, and $z = [\langle e, f\rangle]$. We have

$$\begin{aligned}
(x+y)+z &= ([\langle a,b\rangle] + [\langle c,d\rangle]) + [\langle e,f\rangle] \\
&= [\langle a+c, b+d\rangle] + [\langle e,f\rangle] \\
&= [\langle (a+c)+e, (b+d)+f\rangle] \\
&= [\langle a+(c+e), b+(d+f)\rangle] \\
&= [\langle a,b\rangle] + [\langle c+e, d+f\rangle] \\
&= [\langle a,b\rangle] + ([\langle c,d\rangle] + [\langle e,f\rangle]) \\
&= x+(y+z)
\end{aligned}$$

Where the fourth equality follows from associativity of $\mathbb{N}$. □

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Proof. (2) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z}(x + y = y + x)$$

Let arbitrary $x, y \in \mathbb{Z}$. Then there exists $a, b, c, d \in \mathbb{N}$ such that $x = [\langle a, b\rangle]$ and $y = [\langle c, d\rangle]$. It follows that

$$\begin{aligned} x + y = [\langle a, b\rangle] + [\langle c, d\rangle] = [\langle a + c, b + d\rangle] \\ = [\langle c + a, d + b\rangle] = [\langle c, d\rangle] + [\langle a, b\rangle] = y + x \end{aligned} \quad \square$$

Proof. (3) We want to show

$$\forall x \in \mathbb{Z}(x + \hat{0} = x)$$

Let arbitrary $x \in \mathbb{Z}$. Then there exists $a, b \in \mathbb{N}$ such that $x = [\langle a, b\rangle]$. We have $a + (b + 1) = b + (a + 1)$ and therefore $\langle a, b\rangle \sim \langle a + 1, b + 1\rangle$. So

$$x = [\langle a, b\rangle] = [\langle a + 1, b + 1\rangle] = [\langle a, b\rangle] + [\langle 1, 1\rangle] = x + \hat{0} \quad \square$$

Proof. (4) We want to show

$$\forall x \in \mathbb{Z}(x + (-x) = \hat{0})$$

Let arbitrary $x \in \mathbb{Z}$. Then there exists $a, b \in \mathbb{N}$ such that $x = [\langle a, b\rangle]$. Then by definition $-x = [\langle b, a\rangle]$. We have $x + (-x) = [\langle a, b\rangle] + [\langle b, a\rangle] = [\langle a + b, b + a\rangle]$. We also know $(a + b) + 1 = (b + a) + 1$, meaning $\langle a + b, b + a\rangle \sim \langle 1, 1\rangle$. Putting it all together we have

$$x + (-x) = [\langle a + b, b + a\rangle] = [\langle 1, 1\rangle] = \hat{0} \quad \square$$

Proof. (5) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} \forall z \in \mathbb{Z}((xy)z = x(yz))$$

Let arbitrary $x, y, z \in \mathbb{Z}$. Then there exists $a, b, c, d, e, f \in \mathbb{N}$ such that $x = [\langle a, b\rangle]$, $y = [\langle c, d\rangle]$, and $z = [\langle e, f\rangle]$. We have

$$\begin{aligned} (xy)z &= [\langle ac + bd, ad + bc\rangle] \cdot [\langle e, f\rangle] \\ &= [\langle (ac + bd)e + (ad + bc)f, (ac + bd)f + (ad + bc)e\rangle] \\ &= [\langle ((ac)e + (bd)e) + ((ad)f + (bc)f), ((ac)f + (bd)f) + ((ad)e + (bc)e)\rangle] \\ &= [\langle (a(ce) + a(df)) + (b(de) + b(cf)), (a(cf) + a(de)) + (b(ce) + b(df))\rangle] \\ &= [\langle a(ce + df) + b(cf + de), a(cf + de) + b(ce + df)\rangle] \\ &= [\langle a, b\rangle] \cdot [\langle ce + df, cf + de\rangle] \\ &= [\langle a, b\rangle] \cdot ([\langle c, d\rangle] \cdot [\langle e, f\rangle]) \\ &= x(yz) \end{aligned} \quad \square$$

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Proof. (6) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z}(xy = yx)$$

Let arbitrary $x, y \in \mathbb{Z}$. Then there exists $a, b, c, d \in \mathbb{N}$ such that $x = [\langle a, b\rangle]$ and $y = [\langle c, d\rangle]$. It follows that

$$xy = [\langle ac + bd, ad + bc\rangle] = [\langle ca + db, cb + da\rangle] = [\langle c, d\rangle] \cdot [\langle a, b\rangle] = yx \qquad \square$$

Proof. (7) We want to show

$$\forall x \in \mathbb{Z}(x \cdot \hat{1} = x)$$

Let arbitrary $x \in \mathbb{Z}$. Then there exists $a, b \in \mathbb{N}$ such that $x = [\langle a, b\rangle]$. We have

$$\begin{aligned} x \cdot \hat{1} = [\langle a, b\rangle] \cdot [\langle 1 + 1, 1\rangle] &= [\langle a(1 + 1) + b \cdot 1, a \cdot 1 + b(1 + 1)\rangle] \\ &= [\langle (a + a) + b, a + (b + b)\rangle] \end{aligned}$$

We also have $a + (a + (b + b)) = b + ((a + a) + b)$, meaning $\langle a, b\rangle \sim \langle (a + a) + b, a + (b + b)\rangle$. Combining these two facts we have

$$x \cdot \hat{1} = [\langle (a + a) + b, a + (b + b)\rangle] = [\langle a, b\rangle] = x \qquad \square$$

Proof. (8) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} \forall z \in \mathbb{Z}(x(y + z) = xy + xz)$$

Let arbitrary $x, y, z \in \mathbb{Z}$. Then there exists $a, b, c, d, e, f \in \mathbb{N}$ such that $x = [\langle a, b\rangle]$, $y = [\langle c, d\rangle]$, and $z = [\langle e, f\rangle]$. We have

$$\begin{aligned} x(y + z) &= [\langle a, b\rangle] \cdot [\langle c + e, d + f\rangle] \\ &= [\langle a(c + e) + b(d + f), a(d + f) + b(c + e)\rangle] \\ &= [\langle (ac + bd) + (ae + bf), (ad + bc) + (af + be)\rangle] \\ &= [\langle ac + bd, ad + bc\rangle] + [\langle ae + bf, af + be\rangle] \\ &= [\langle a, b\rangle] \cdot [\langle c, d\rangle] + [\langle a, b\rangle] \cdot [\langle e, f\rangle] \\ &= xy + xz \end{aligned} \qquad \square$$

(next page)

Lemma 1.25. $\forall a \in \mathbb{N} \forall b \in \mathbb{N}([\langle a, b\rangle] = \hat{0} \leftrightarrow a = b)$

Proof. Let arbitrary $a, b \in \mathbb{N}$. First suppose that $[\langle a, b\rangle] = \hat{0} = [\langle 1, 1\rangle]$. Then $a+1 = b+1$ and by cancellation $a = b$. Next suppose $a = b$. Then $a+1 = b+1$ and $\langle a, b\rangle \sim \langle 1, 1\rangle$. We then have $[\langle a, b\rangle] = [\langle 1, 1\rangle] = \hat{0}$. □

Proof. (9) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z}(xy = \hat{0} \to (x = \hat{0} \vee y = \hat{0}))$$

Let arbitrary $x, y \in \mathbb{Z}$ and suppose that $xy = \hat{0}$. The statement after the implication is logically equivalent to $x \neq \hat{0} \to y = \hat{0}$. Suppose $x \neq \hat{0}$. There exists $a, b, c, d \in \mathbb{N}$ such that $x = [\langle a, b\rangle]$ and $y = [\langle c, d\rangle]$. We then have $xy = [\langle ac + bd, ad + bc\rangle] = \hat{0}$; by the previous lemma $ac + bd = ad + bc$. That lemma also implies $a \neq b$; so by trichotomy of the natural numbers we have $a < b$ or $b < a$.

In the first case suppose $a < b$. Then there exists $g \in \mathbb{N}$ such that $a + g = b$. We then have

$$(ac + ad) + gd = ac + (a + g)d = ad + (a + g)c = (ac + ad) + gc$$

Cancellation of addition on the naturals allows us to conclude that $gd = gc$. Cancellation of multiplication allows us to conclude $c = d$. By the previous lemma this implies $y = [\langle c, d\rangle] = \hat{0}$.

In the other case suppose $b < a$. Then there exists $g \in \mathbb{N}$ such that $b+g = a$. We then have

$$(bc + bd) + gc = (b + g)c + bd = (b + g)d + bc = (bc + bd) + gd$$

Similarly again we conclude $gc = gd$ and $c = d$, which then implies $y = [\langle c, d\rangle] = \hat{0}$. □

(next page)

Proof. (10) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z}((x < y \vee x = y \vee y < x) \wedge \\ \neg(x < y \wedge x = y) \wedge \neg(x < y \wedge y < x) \wedge \neg(x = y \wedge y < x))$$

Let arbitrary $x, y \in \mathbb{Z}$. Then there exists $a, b, c, d \in \mathbb{N}$ such that $x = [\langle a, b\rangle]$ and $y = [\langle c, d\rangle]$. We start with the latter three statements. First suppose $x < y$ and $x = y$. By lemma 1.21 $a + d < b + c$ and by definition $a + d = b + c$; this contradicts trichotomy for the naturals. Next suppose $x < y$ and $y < x$. Again this then implies $a + d < b + c$ and $b + c < a + d$ which contradicts trichotomy. Supposing $x = y$ and $y < x$ also contradicts trichotomy in a similar manner.

Finally the former statement. We know that $a + d, b + c \in \mathbb{N}$. By trichotomy we must have

$$a + d < b + c \vee a + d = b + c \vee b + c < a + d$$

which by lemma 1.21 and the definition of equality means

$$[\langle a, b\rangle] < [\langle c, d\rangle] \vee [\langle a, b\rangle] = [\langle c, d\rangle] \vee [\langle c, d\rangle] < [\langle a, b\rangle]$$

which is equivalent to our desired result. □

Proof. (11) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} \forall z \in \mathbb{Z}(x < y \wedge y < z \to x < z)$$

Let arbitrary $x, y, z \in \mathbb{Z}$. Suppose $x < y$ and $y < z$. Then by definition there exists $a, b, c, d, e, f \in \mathbb{N}$ such that $x = [\langle a, b\rangle], y = [\langle c, d\rangle]$, and $z = [\langle e, f\rangle]$. By lemma 1.21 we can assert that $a + d < b + c$ and $c + f < d + e$. By definition there then exists $g, g' \in \mathbb{N}$ such that $(a + d) + g = b + c$ and $(c + f) + g' = d + e$. The former, along with cancellation, associativity and commutativity, allows us to assert

$$((a + d) + g) + f = ((a + d) + f) + g = (b + c) + f$$

So $(a + d) + f < (b + c) + f$. Similarly the latter allows us to assert

$$((c + f) + g') + b = ((b + c) + f) + g' = (d + e) + b = b + (d + e)$$

So $(b + c) + f < b + (d + e)$. By transitivity of the naturals we then have

$$(a + d) + f < b + (d + e)$$

Then there exists $g_0 \in \mathbb{N}$ such that $((a + d) + f) + g_0 = b + (d + e)$. So we have

$$d + ((a + f) + g_0) = ((a + d) + f) + g_0 = b + (d + e) = d + (b + e)$$

which by cancellation means

$$(a + f) + g_0 = b + e$$

and $a + f < b + e$. By lemma 1.21 this means $[\langle a, b\rangle] < [\langle e, f\rangle]$ and $x < z$. □

(next page)

Proof. (12) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} \forall z \in \mathbb{Z}(x < y \to x + z < y + z)$$

Let arbitrary $x, y, z \in \mathbb{Z}$. By definition there exists $a, b, c, d, e, f \in \mathbb{N}$ such that $x = [\langle a, b\rangle], y = [\langle c, d\rangle]$, and $z = [\langle e, f\rangle]$. Suppose $x < y$. By lemma 1.21 we then have $a+d < b+c$. By theorem 1.10 part 4 and commutativity and associativity of the naturals, we can assert

$$(a + e) + (d + f) = (a + d) + (e + f) < (b + c) + (e + f) = (b + f) + (c + e)$$

which by lemma 1.21 and the definition of addition, implies

$$[\langle a, b\rangle] + [\langle e, f\rangle] = [\langle a + e, b + f\rangle] < [\langle c + e, d + f\rangle] = [\langle c, d\rangle] + [\langle e, f\rangle]$$

which is equivalent to our desired result. $\square$

Proof. (13) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} \forall z \in \mathbb{Z}(x < y \land z > \hat{0} \to xz < yz)$$

Let arbitrary $x, y, z \in \mathbb{Z}$. By definition there exists $a, b, c, d, e, f \in \mathbb{N}$ such that $x = [\langle a, b\rangle], y = [\langle c, d\rangle]$, and $z = [\langle e, f\rangle]$. Suppose that $x < y$ and $z > \hat{0}$. By lemma 1.21 we have $a + d < b + c$ and $1 + f < 1 + e$. By theorem 1.10 part 4 we have $f < e$, so there exists $k \in \mathbb{N}$ such that $f + k = e$. By theorem 1.10 part 6 we have $k(a + d) < k(b + c)$. Part 4 of the same theorem allows us to then assert

$$k(a + d) + ((a + d)f + (b + c)f) < k(b + c) + ((a + d)f + (b + c)f)$$

The left hand side is equivalent to

$$\begin{aligned}
&k(a + d) + ((a + d)f + (b + c)f) \\
&= (ka + kd) + ((af + df) + (bf + cf)) \\
&= ((ka + kd) + (af + df)) + (bf + cf) \\
&= (a(k + f) + d(k + f)) + (bf + cf) \\
&= (ae + de) + (bf + cf) \\
&= (ae + bf) + (cf + de)
\end{aligned}$$

Similarly the right hand side is equivalent to

$$\begin{aligned}
&k(b + c) + ((a + d)f + (b + c)f) \\
&= (kb + kc) + ((af + df) + (bf + cf)) \\
&= (b(k + f) + c(k + f)) + (af + df) \\
&= (be + ce) + (af + df) \\
&= (af + be) + (ce + df)
\end{aligned}$$

(next page)

We can then conclude that $(ae+bf)+(cf+de)<(af+be)+(ce+df)$. By lemma 1.21 and the definition of multiplication on the integers we then have

$$[\langle a,b\rangle]\cdot[\langle e,f\rangle]=[\langle ae+bf,af+be\rangle]<[\langle ce+df,cf+de\rangle]=[\langle c,d\rangle]\cdot[\langle e,f\rangle]$$

Which is equivalent to our desired conclusion. $\square$

Proof. (14) We want to show $\hat{0}\neq\hat{1}$. Suppose for contradiction that they are equal, meaning $[\langle 1,1\rangle]=[\langle 1{+}1,1\rangle]$. By definition we then have $1{+}1=1{+}(1{+}1)$, which by cancellation implies $1=1+1$, contradicting part 5 of theorem 1.7. $\square$

1.2.5 Joining the naturals and the integers

Definition 1.26. *Let $x \in \mathbb{Z}$. x is **positive** if $x > \hat{0}$. x is **negative** if $x < \hat{0}$.*

Theorem 1.27. *Let $i : \mathbb{N} \to \mathbb{Z}$ be defined by $i(n) = [\langle n+1, 1\rangle]$ for all $n \in \mathbb{N}$.*

1. *The function $i : \mathbb{N} \to \mathbb{Z}$ is injective.*
2. *$i(\mathbb{N}) = \{x \in \mathbb{Z} \mid x > \hat{0}\}$*
3. *$i(1) = \hat{1}$*
4. *Let $a, b \in \mathbb{N}$. Then*
   (a) *$i(a+b) = i(a) + i(b)$*
   (b) *$i(ab) = i(a)i(b)$*
   (c) *$a < b$ if and only if $i(a) < i(b)$*

Proof. (1) Defining i as the set such that

$$\forall x(x \in i \leftrightarrow \exists a \in \mathbb{N} \exists b(x = \langle a, b\rangle \wedge b = [\langle a+1, 1\rangle]))$$

Existence of such a set can be shown using the subset axiom, with $\mathbb{N} \times \mathbb{Z}$ as a bounding set. We start by showing functionality. Let arbitrary $a \in \mathrm{dom}(i)$. By definition there exists b such that $\langle a, b\rangle \in i$; as defined this implies $a \in \mathbb{N}$ and $b = [\langle a+1, 1\rangle]$. Let b' be arbitrary such that $\langle a, b'\rangle \in i$. Again by definition we immediately have $b' = [\langle a+1, 1\rangle] = b$.

Next the domain bound $\mathrm{dom}(i) = \mathbb{N}$. Let a be arbitrary and suppose that $a \in \mathrm{dom}(i)$. By definition there then exists b such that $\langle a, b\rangle \in i$ and it immediately follows that $a \in \mathbb{N}$. Now suppose $a \in \mathbb{N}$. By definition $[\langle a+1, 1\rangle] \in \mathbb{Z}$ and $\langle a, [\langle a+1, 1\rangle]\rangle \in i$. So $a \in \mathrm{dom}(i)$.

Next the range bound $\mathrm{ran}(i) \subseteq \mathbb{Z}$. Let arbitrary $b \in \mathrm{ran}(i)$. Then by definition there exists a such that $\langle a, b\rangle \in i$. It immediately follows that $a \in \mathbb{N}$ and $b = [\langle a+1, 1\rangle] \in \mathbb{Z}$.

Finally we show injectivity. Let arbitrary $b \in \mathrm{ran}(i)$. By definition there exists a such that $\langle a, b\rangle \in i$; it also follows that $a \in \mathbb{N}$ and $b = [\langle a+1, 1\rangle]$. Let a' be arbitrary and suppose $\langle a', b\rangle \in i$; similarly we have $a' \in \mathbb{N}$ and $b = [\langle a'+1, 1\rangle]$. So $[\langle a+1, 1\rangle] = [\langle a'+1, 1\rangle]$, which implies $\langle a+1, 1\rangle \sim \langle a'+1, 1\rangle$ and $(a+1)+1 = 1+(a'+1)$. By commutativity and cancellation we then have $a = a'$. □

(next page)

Proof. (2) We show $\{i(n) \mid n \in \mathbb{N}\} = \{x \in \mathbb{Z} \mid x > \hat{0}\}$. More specifically, denoting the former as A and the latter B, they are the sets such that

$$\forall x(x \in A \leftrightarrow \exists n \in \mathbb{N}(x = i(n)))$$
$$\forall x(x \in B \leftrightarrow x \in \mathbb{Z} \wedge x > \hat{0})$$

Existence of A can be shown using the subset axiom with $\mathbb{Z}$ as a bounding set. Existence of B is immediately assured by the subset axiom. We want to show $A = B$.

Let x be arbitrary and suppose $x \in A$. Then there exists $n \in \mathbb{N}$ such that $x = i(n)$. So $\langle n, x\rangle \in i$ and $x \in \operatorname{ran}(i) \subseteq \mathbb{Z}$. By definition $x = [\langle n+1, 1\rangle]$. Since we know $(1+1)+n = 1+(n+1)$ it follows that $1+1 < 1+(n+1)$ and by lemma 1.21 that $\hat{0} = [\langle 1, 1\rangle] < [\langle n+1, 1\rangle] = x$. So $x \in B$.

Now suppose $x \in B$. Then $x \in \mathbb{Z}$ and $x > \hat{0}$. By definition there exists $a, b \in \mathbb{N}$ such that $x = [\langle a, b\rangle] > \hat{0}$. By lemma 1.21 $1+b < 1+a$, so there exists $k \in \mathbb{N}$ such that $(1+b)+k = 1+a$; associativity and cancellation allows us to conclude that $b+k = a$ it is apparent that $[\langle k+1, 1\rangle] \in \mathbb{Z}$ and that by definition $\langle k, [\langle k+1, 1\rangle]\rangle \in i$. We know $(1+b)+k = 1+a$; so $(k+1)+b = 1+a$, $\langle k+1, 1\rangle \sim \langle a, b\rangle$, and $x = [\langle a, b\rangle] = [\langle k+1, 1\rangle] = i(k)$, meaning $x \in A$. $\square$

Proof. (3) We defined i such that

$$\forall x(x \in i \leftrightarrow \exists a \in \mathbb{N} \exists b(x = \langle a, b\rangle \wedge b = [\langle a+1, 1\rangle]))$$

By definition we have $\langle 1, [\langle 1+1, 1\rangle]\rangle \in i$. So $i(1) = [\langle 1+1, 1\rangle] = \hat{1}$. $\square$

Proof. (4a) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(i(a+b) = i(a) + i(b))$$

Let arbitrary $a, b \in \mathbb{N}$. See that $\langle ((a+b)+1)+1, 1+1\rangle \sim \langle (a+b)+1, 1\rangle$ since

$$(((a+b)+1)+1)+1 = (1+1)+((a+b)+1)$$

We therefore have the following string of equivalences

$$\begin{aligned} i(a+b) &= [\langle (a+b)+1, 1\rangle] = [\langle ((a+b)+1)+1, 1+1\rangle] \\ &= [\langle (a+1)+(b+1), 1+1\rangle] = [\langle a+1, 1\rangle] + [\langle b+1, 1\rangle] \\ &= i(a) + i(b) \end{aligned}$$ $\square$

(next page)

Proof. (4b) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(i(ab) = i(a)i(b))$$

Letting arbitrary $a, b \in \mathbb{N}$, see that $\langle (a+1)(b+1)+1, (a+1)+(b+1) \rangle \sim \langle ab+1, 1 \rangle$:

$$\begin{aligned} ((a+1)(b+1)+1)+1 &= (((a+1)b+(a+1))+1)+1 \\ &= (((ab+b)+(a+1))+1)+1 \\ &= ((a+1)+(b+1))+(ab+1) \end{aligned}$$

We therefore have

$$i(ab) = [\langle ab+1, 1 \rangle] = [\langle (a+1)(b+1)+1, (a+1)+(b+1) \rangle] = i(a)i(b) \quad \square$$

Proof. (4c) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(a < b \leftrightarrow i(a) < i(b))$$

Let arbitrary $a, b \in \mathbb{N}$. First suppose $a < b$. Then by theorem 1.10 part 4 we know $a+1 < b+1$; applying the same theorem again we have $(a+1)+1 < (b+1)+1 = 1+(b+1)$. By lemma 1.21 this implies $[\langle a+1, 1 \rangle] < [\langle b+1, 1 \rangle]$ and $i(a) < i(b)$.

Now instead suppose $i(a) < i(b)$. This implies $[\langle a+1, 1 \rangle] < [\langle b+1, 1 \rangle]$. By lemma 1.21 we then have $(a+1)+1 < 1+(b+1) = (b+1)+1$. Applying theorem 1.10 part 4 twice we then have $a < b$. $\square$

1.2.6 More properties

(Denoting $\hat{0}$ as 0 and $\hat{1}$ as 1)

Theorem 1.28. *Let $x, y, z \in \mathbb{Z}$.*

1. *If $x + z = y + z$, then $x = y$ (Cancellation Law for Addition)*
2. $-(-x) = x$
3. $-(x + y) = (-x) + (-y)$
4. $x \cdot 0 = 0$
5. $(-x)y = -xy = x(-y)$
6. *If $z \neq 0$ and $xz = yz$, then $x = y$ (Cancellation Law for Multiplication)*
7. *$x > 0$ if and only if $-x < 0$ and $x < 0$ if and only if $-x > 0$*
8. $0 < 1$
9. *If $x \leq y$ and $y \leq x$. Then $x = y$*
10. *If $x > 0$ and $y > 0$, then $xy > 0$. If $x > 0$ and $y < 0$, then $xy < 0$.*

Proof. (1) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} \forall z \in \mathbb{Z}(x + z = y + z \rightarrow x = y)$$

Let arbitrary $x, y, z \in \mathbb{Z}$. Suppose that $x + z = y + z$. The desired result follows from several parts of theorem 1.24:

$$\begin{aligned} x &\overset{(3)}{=} x + 0 \overset{(4)}{=} x + (z + (-z)) \overset{(1)}{=} (x + z) + (-z) = (y + z) + (-z) \\ &\overset{(1)}{=} y + (z + (-z)) \overset{(4)}{=} y + 0 \overset{(3)}{=} y \end{aligned}$$

□

Proof. (2) We want to show

$$\forall x \in \mathbb{Z}(-(-x) = x)$$

Let arbitrary $x \in \mathbb{Z}$. Part 4 of theorem 1.24 allows us to state

$$(-x) + (-(-x)) = 0$$

We then have, by the same theorem,

$$\begin{aligned} x &\overset{(3)}{=} x + 0 = x + ((-x) + (-(-x))) \overset{(1)}{=} (x + (-x)) + (-(-x)) \\ &\overset{(4)}{=} 0 + (-(-x)) \overset{(2,3)}{=} (-(-x)) \end{aligned}$$

□

(next page)

Proof. (3) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z}(-(x+y) = (-x) + (-y))$$

Let arbitrary $x, y \in \mathbb{Z}$. By part 4 of theorem 1.24 we have

$$(x+y) + (-(x+y)) = 0$$

We then have, by the same theorem,

$$\begin{aligned}
(-x) + (-y) &\overset{(3)}{=} ((-x) + (-y)) + 0 = ((-x) + (-y)) + ((x+y) + (-(x+y))) \\
&\overset{(1)}{=} (((-x) + (-y)) + (x+y)) + (-(x+y)) \\
&\overset{(1,2)}{=} ((-x) + (((-y) + y) + x)) + (-(x+y)) \\
&\overset{(4)}{=} ((-x) + (0 + x)) + (-(x+y)) \\
&\overset{(2,3)}{=} ((-x) + x) + (-(x+y)) \\
&\overset{(2,4)}{=} 0 + (-(x+y)) \\
&\overset{(2,3)}{=} -(x+y)
\end{aligned}$$

□

Proof. (4) We want to show

$$\forall x \in \mathbb{Z}(x \cdot 0 = 0)$$

Let arbitrary $x \in \mathbb{Z}$. We have, by theorem 1.24,

$$x \cdot 0 + 0 \overset{(3)}{=} x \cdot 0 \overset{(3)}{=} x(0+0) \overset{(8)}{=} x \cdot 0 + x \cdot 0$$

By commutativity and cancellation we have $0 = x \cdot 0$. □

Proof. (5) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z}((-x)y = -(xy) = x(-y))$$

Let arbitrary $x, y \in \mathbb{Z}$. We first show $(-x)y = -(xy)$. By part 4 of this theorem and several parts of theorem 1.24 we have

$$xy + (-(xy)) \overset{(4)}{=} 0 = y \cdot 0 \overset{(4)}{=} y(x + (-x)) \overset{(8)}{=} yx + y(-x) \overset{(6)}{=} xy + (-x)y$$

where cancellation on addition immediately yields $-(xy) = (-x)y$. Similarly we have

$$xy + (-(xy)) \overset{(4)}{=} 0 = x \cdot 0 \overset{(4)}{=} x(y + (-y)) \overset{(8)}{=} xy + x(-y)$$

where it follows that $-(xy) = x(-y)$. □

(next page)

Proof. (6) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} \forall z \in \mathbb{Z}((z \neq 0 \land xz = yz) \to x = y)$$

Let arbitrary $x, y, z \in \mathbb{Z}$. Suppose that $z \neq 0$ and $xz = yz$. By part 5 of this theorem and several parts of theorem 1.24 we have

$$0 \overset{(4)}{=} xz + (-(xz)) = yz + (-(xz)) = yz + (-x)z \overset{(6,8)}{=} (y + (-x))z$$

Part 9 of theorem 1.24 then allows us to conclude $y + (-x) = 0$ or $z = 0$. By our earlier assumptions we must then have $y + (-x) = 0$. So by part 4 of theorem 1.24,

$$y + (-x) = 0 = x + (-x)$$

and by cancellation on addition $y = x$. $\square$

Proof. (7) We want to show

$$\forall x \in \mathbb{Z}((x > 0 \leftrightarrow -x < 0) \land (x < 0 \leftrightarrow -x > 0))$$

Let arbitrary $x \in \mathbb{Z}$. First suppose $x > 0$. It follows from theorem 1.24 that

$$-x \overset{(3,2)}{=} 0 + (-x) \overset{(12)}{<} x + (-x) \overset{(4)}{=} 0$$

Next suppose $-x < 0$. Similarly,

$$0 \overset{(4)}{=} x + (-x) \overset{(2,12)}{<} x + 0 \overset{(3)}{=} x$$

Next suppose $x < 0$. We have

$$-x \overset{(3,2)}{=} 0 + (-x) \overset{(12)}{>} x + (-x) \overset{(4)}{=} 0$$

Finally suppose $-x > 0$. We have

$$0 \overset{(4)}{=} x + (-x) \overset{(2,12)}{>} x + 0 \overset{(3)}{=} x$$ $\square$

(next page)

Proof. (8) We want to show $0 < 1$. We know by theorem 1.24 part 14 that $0 \neq 1$. So by part 10 (trichotomy) of that same theorem we either have $0 < 1$ or $0 > 1$. Suppose for contradiction that the latter is true. By part 7 of this theorem we immediately have $0 < -1$. By part 13 of theorem 1.24 we then have $0(-1) < (-1)(-1)$. By an earlier part of this theorem and theorem 1.24,

$$0 = 0(-1) < (-1)(-1) = 1(-(-1)) = 1 \cdot 1 = 1$$

Which contradicts trichotomy since we assumed $1 < 0$. We therefore must have $0 < 1$. ☐

Proof. (9) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z}((x \leq y \wedge y \leq x) \to x = y)$$

Let arbitrary $x, y \in \mathbb{Z}$. Suppose that $x \leq y$ and $y \leq x$. Lemma 1.23 implies that $x < y \vee x = y$ and $y < x \vee y = x$. Supposing $x < y$, $y < x \vee y = x$ is a contradiction in either case. Therefore $x < y \vee x = y$ implies that it must be the case that $x = y$. ☐

Proof. (10) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z}(([x > 0 \wedge y > 0] \to xy > 0) \wedge ([x > 0 \wedge y < 0] \to xy < 0))$$

Let $x, y \in \mathbb{Z}$ be arbitrary. First suppose $x > 0$ and $y > 0$. By theorem 1.24 part 13 we immediately have

$$xy > 0 \cdot y = 0$$

Where the equality follows from commutativity and part 4 of this theorem. Next suppose $x > 0$ and $y < 0$. Similarly we have

$$xy = yx < 0 \cdot x = 0$$ ☐

1.2.7 Discreteness

Theorem 1.29. *Let $x \in \mathbb{Z}$. Then there is no $y \in \mathbb{Z}$ such that $x < y < x + 1$.*

Proof. We want to show

$$\forall x \in \mathbb{Z}(\neg \exists y \in \mathbb{Z}(x < y < x + 1))$$

Let arbitrary $x \in \mathbb{Z}$. Suppose for contradiction that there exists some $y \in \mathbb{Z}$ such that $x < y < x + 1$. By theorem 1.24 we have

$$\begin{aligned} 1 = x + ((-x) + 1) < y + ((-x) + 1) <&(x + 1) + ((-x) + 1) \\ &= (1 + (x + (-x))) + 1 = 1 + 1 \end{aligned}$$

By theorem 1.28 part 8 and transitivity we have $0 < y + ((-x) + 1)$. It then follows from theorem 1.27 that there exists $n \in \mathbb{N}$ such that $i(n) = y+((-x)+1)$, where $i : \mathbb{N} \to \mathbb{Z}$. By the same theorem we have

$$i(1) = 1 < i(n) < 1 + 1 = i(1) + i(1) = i(1 + 1)$$

and

$$1 < n < 1 + 1$$

Which contradicts part 9 of theorem 1.10. □

Lemma 1.30. $\forall x \in \mathbb{Z} \forall y \in \mathbb{Z}(x > y \to x \geq y + 1)$

Proof. Let $x, y \in \mathbb{Z}$ be arbitrary. Suppose $x > y$. Suppose for contradiction that $x < y + 1$. This contradicts theorem 1.29 since $y < x < y + 1$. By trichotomy we must therefore $x \geq y + 1$. □

(next page)

Theorem 1.31. *Let $x, y \in \mathbb{Z}$. $xy = 1$ iff $x = 1 = y$ or $x = -1 = y$.*

Proof. We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z}(xy = 1 \leftrightarrow (x = 1 = y \vee x = -1 = y))$$

Let arbitrary $x, y \in \mathbb{Z}$. Suppose that $xy = 1$. It follows from theorem 1.24 that $xy \neq 0$. By contradiction it can then be shown that $x \neq 0$ and $y \neq 0$. By trichotomy we must have either $x > 0$ or $x < 0$.

First consider the case where $x > 0$. Since $xy = 1 > 0$, by trichotomy on y it can be shown by contradiction that $y > 0$; and by the previous lemma $y \geq 1$. Suppose $x \neq 1$; by trichotomy we have either $x < 1$ or $x > 1$. The first case implies $0 < x < 1$, which contradicts theorem 1.29. In the second case by theorem 1.24 we would have $xy > y \geq 1$, which implies $xy > 1$, contradicting trichotomy. We must therefore have $x = 1$. It immediately follows that $1 = xy = 1 \cdot y = y$.

Next consider the case where $x < 0$. By trichotomy $y < 0 \vee y = 0 \vee y > 0$. The latter two cases are contradictions since $y = 0$ would imply $xy = 0 \neq 1$ and $y > 0$ would, by theorem 1.28, imply $xy < 0 < 1$. We therefore conclude that $y < 0$. Again by theorem 1.28, we have $-x > 0$ and $-y > 0$ and $(-x)(-y) = xy = 1$. Using an identical strategy to the previous case, it follows that $(-x) = 1 = (-y)$ and $x = -1 = y$.

Now for the reverse implication. Supposing $x = 1 = y$ it immediately follows that $xy = 1 \cdot 1 = 1$. Supposing $x = -1 = y$ we have $xy = (-1)(-1) = 1 \cdot 1 = 1$. $\square$

1.3 Construction of the Rationals

1.3.1 Construction

Definition 1.32. *Let $\mathbb{Z}^* = \mathbb{Z} - \{0\}$. The relation $\asymp$ on $\mathbb{Z} \times \mathbb{Z}^*$ is defined by $(x, y) \asymp (z, w)$ if and only if $xw = yz$, for all $(x, y), (z, w) \in \mathbb{Z} \times \mathbb{Z}^*$. That is, the set such that*

$$\forall a(a \in \asymp \leftrightarrow \exists \langle x, y\rangle \in \mathbb{Z} \times \mathbb{Z}^* \exists \langle z, w\rangle \in \mathbb{Z} \times \mathbb{Z}^*(a = \langle\langle x, y\rangle, \langle z, w\rangle\rangle \wedge xw = yz))$$

(Existence follows from using $(\mathbb{Z} \times \mathbb{Z}^*) \times (\mathbb{Z} \times \mathbb{Z}^*)$ as a bounding set.)

Lemma 1.33. *The relation $\asymp$ is an equivalence relation on $\mathbb{Z} \times \mathbb{Z}^*$.*

Proof. Showing $\asymp \subseteq (\mathbb{Z} \times \mathbb{Z}^*) \times (\mathbb{Z} \times \mathbb{Z}^*)$ is straightforward. We start with reflexivity on $\mathbb{Z} \times \mathbb{Z}^*$. Let arbitrary $x \in \mathbb{Z} \times \mathbb{Z}^*$; there exists $\langle a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = \langle a, b\rangle$. Since $\mathbb{Z}^* \subseteq \mathbb{Z}$, $a, b \in \mathbb{Z}$, so $ab = ba$. It then follows by definition that $\langle x, x\rangle = \langle\langle a, b\rangle, \langle a, b\rangle\rangle \in \asymp$.

Now for symmetry. Let a, b be arbitrary such that $\langle a, b\rangle \in \asymp$. By definition there then exists $\langle x, y\rangle, \langle z, w\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $a = \langle x, y\rangle$, $b = \langle z, w\rangle$, and $xw = yz$. By commutativity we have $zy = wx$, meaning $\langle b, a\rangle = \langle\langle z, w\rangle, \langle x, y\rangle\rangle \in \asymp$.

Finally transitivity. Let a, b, c be arbitrary such that $\langle a, b\rangle, \langle b, c\rangle \in \asymp$. By definition there exists $\langle x, y\rangle, \langle z, w\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ where $a = \langle x, y\rangle$, $b = \langle z, w\rangle$, and $xw = yz$. There also exists $\langle x', y'\rangle, \langle z', w'\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ where $b = \langle x', y'\rangle$, $c = \langle z', w'\rangle$, and $x'w' = y'z'$. It follows that $\langle z, w\rangle = \langle x', y'\rangle$ and $zw' = wz'$. We then have $(xw)w' = (yz)w'$ and $y(zw') = y(wz')$. By associativity we then have $(xw)w' = (yz)w' = y(zw') = y(wz')$. By commutativity and associativity it follows that $(xw')w = (yz')w$. Since $w \in \mathbb{Z}^*$, $w \neq 0$ and by cancellation we have $xw' = yz'$. By definition $\langle a, c\rangle = \langle\langle x, y\rangle, \langle z', w'\rangle\rangle \in \asymp$. $\square$

Definition 1.34. *The set of* ***rational numbers****, denoted $\mathbb{Q}$, is the quotient set $(\mathbb{Z} \times \mathbb{Z}^*)/\asymp$. That is,*

$$\forall x(x \in \mathbb{Q} \leftrightarrow \exists \langle a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^*(x = [\langle a, b\rangle]))$$

where

$$\forall \langle a, b\rangle \forall x(x \in [\langle a, b\rangle] \leftrightarrow \langle a, b\rangle \asymp x)$$

Existence of each equivalence class can be shown using $\mathrm{ran}(\asymp)$ as a bounding set. The same can be done for $\mathbb{Q}$ using $\mathscr{P}(\mathrm{ran}(\asymp))$.

Definition 1.35. *The elements $\bar{0}, \bar{1} \in \mathbb{Q}$ are defined by $\bar{0} = [\langle 0, 1\rangle]$ and $\bar{1} = [\langle 1, 1\rangle]$.*

1.3.2 Operations

Definition 1.36. *The binary operation $+$ on $\mathbb{Q}$ is defined as $+ : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$ where*

$$\forall x(x \in + \leftrightarrow \exists \langle a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^* \exists \langle c, d\rangle \in \mathbb{Z} \times \mathbb{Z}^* (x = \langle\langle[\langle a, b\rangle], [\langle c, d\rangle]\rangle, [\langle ad + bc, bd\rangle]\rangle))$$

(Existence follows from using $(\mathbb{Q} \times \mathbb{Q}) \times \mathbb{Q}$ as a bounding set.)

Proof. (Domain/Range bounds, Well-definedness, Functionality) Staring with the domain bound, $\text{dom}(+) = \mathbb{Q} \times \mathbb{Q}$. First let arbitrary $x \in \text{dom}(+)$. By definition there exists y such that $\langle x, y\rangle \in +$. By definition there then exists $\langle a, b\rangle, \langle c, d\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = \langle[\langle a, b\rangle], [\langle c, d\rangle]\rangle \in \mathbb{Q} \times \mathbb{Q}$. Now let arbitrary $\langle x, y\rangle \in \mathbb{Q} \times \mathbb{Q}$. By definition there then exists $\langle a, b\rangle, \langle c, d\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b\rangle]$ and $y = [\langle c, d\rangle]$. We then have $\langle\langle[\langle a, b\rangle], [\langle c, d\rangle]\rangle, [\langle ad + bc, bd\rangle]\rangle \in +$. So $\langle x, y\rangle = \langle[\langle a, b\rangle], [\langle c, d\rangle]\rangle \in \text{dom}(+)$.

Next the range bound $\text{ran}(+) \subseteq \mathbb{Q}$. Let arbitrary $y \in \text{ran}(+)$. Then there exists x such that $\langle x, y\rangle \in +$. There then exists $\langle a, b\rangle, \langle c, d\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $y = [\langle ad + bc, bd\rangle]$. We know that $b, d \in \mathbb{Z}^*$. By theorem 1.24 since $b \neq 0$ and $d \neq 0$ we must have $bd \neq 0$. So $bd \in \mathbb{Z} - \{0\} = \mathbb{Z}^*$, and $\langle ad + bc, bd\rangle \in \mathbb{Z} \times \mathbb{Z}^*$. So $y = [\langle ad + bc, bd\rangle] \in \mathbb{Q}$.

Next well-definedness. That is,

$$\begin{aligned}\forall \langle a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^* \forall \langle c, d\rangle \in \mathbb{Z} \times \mathbb{Z}^* \forall \langle a', b'\rangle \in \mathbb{Z} \times \mathbb{Z}^* \forall \langle c', d'\rangle \in \mathbb{Z} \times \mathbb{Z}^* (\\ ([\langle a, b\rangle] = [\langle a', b'\rangle] \wedge [\langle c, d\rangle] = [\langle c', d'\rangle])\\ \to [\langle ad + bc, bd\rangle] = [\langle a'd' + b'c', b'd'\rangle])\end{aligned}$$

Let arbitrary $\langle a, b\rangle, \langle c, d\rangle, \langle a', b'\rangle, \langle c', d'\rangle \in \mathbb{Z} \times \mathbb{Z}^*$. Suppose $[\langle a, b\rangle] = [\langle a', b'\rangle]$ and $[\langle c, d\rangle] = [\langle c', d'\rangle]$. We then have $\langle a, b\rangle \asymp \langle a', b'\rangle$ and $\langle c, d\rangle \asymp \langle c', d'\rangle$; so $ab' = ba'$ and $cd' = dc'$. We then have

$$\begin{aligned}(ad + bc)(b'd') &= (ad)(b'd') + (bc)(b'd')\\ &= (ab')(dd') + (cd')(bb')\\ &= (ba')(dd') + (dc')(bb')\\ &= (bd)(a'd') + (bd)(b'c')\\ &= (bd)(a'd' + b'c')\end{aligned}$$

Which implies $\langle ad + bc, bd\rangle \asymp \langle a'd' + b'c', b'd'\rangle$ and $[\langle ad + bc, bd\rangle] = [\langle a'd' + b'c', b'd'\rangle]$.
(next page)

Finally functionality. Let arbitrary $x \in \mathrm{dom}(+)$. There then exists y such that $\langle x, y\rangle \in +$. By definition there then exists $\langle a, b\rangle, \langle c, d\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = \langle[\langle a, b\rangle], [\langle c, d\rangle]\rangle$ and $y = [\langle ad + bc, bd\rangle]$. Now let y' be arbitrary such that $\langle x, y'\rangle \in +$. Then there exists $\langle a', b'\rangle, \langle c', d'\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = \langle[\langle a', b'\rangle], [\langle c', d'\rangle]\rangle$ and $y' = [\langle a'd' + b'c', b'd'\rangle]$. Since this implies $[\langle a, b\rangle] = [\langle a', b'\rangle]$ and $[\langle c, d\rangle] = [\langle c', d'\rangle]$, by well definedness we have $y = [\langle ad + bc, bd\rangle] = [\langle a'd' + b'c', b'd'\rangle] = y'$. $\square$

Definition 1.37. *The binary operation $\cdot$ on $\mathbb{Q}$ is defined as $\cdot : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$ where*

$$\forall x(x \in \cdot \leftrightarrow \exists\langle a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^* \exists\langle c, d\rangle \in \mathbb{Z} \times \mathbb{Z}^*(x = \langle\langle[\langle a, b\rangle], [\langle c, d\rangle]\rangle, [\langle ac, bd\rangle]\rangle))$$

(Existence follows from using $(\mathbb{Q} \times \mathbb{Q}) \times \mathbb{Q}$ as a bounding set.)

Proof. (Domain/Range bounds, Well-definedness, Functionality) First the domain bound $\mathrm{dom}(\cdot) = \mathbb{Q} \times \mathbb{Q}$. First let arbitrary $x \in \mathrm{dom}(\cdot)$. Then there exists y such that $\langle x, y\rangle \in \cdot$, and $\langle a, b\rangle, \langle c, d\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $\langle x, y\rangle = \langle[\langle a, b\rangle], [\langle c, d\rangle]\rangle \in \mathbb{Q} \times \mathbb{Q}$. Next let arbitary $\langle x, y\rangle \in \mathbb{Q} \times \mathbb{Q}$. There then exists $\langle a, b\rangle, \langle c, d\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b\rangle]$ and $y = [\langle c, d\rangle]$. By definition $\langle\langle[\langle a, b\rangle], [\langle c, d\rangle]\rangle, [\langle ac, bd\rangle]\rangle \in \cdot$; so $\langle x, y\rangle = \langle[\langle a, b\rangle], [\langle c, d\rangle]\rangle \in \mathrm{dom}(\cdot)$.

Next the range bound $\mathrm{ran}(\cdot) \subseteq \mathbb{Q}$.

$\square$